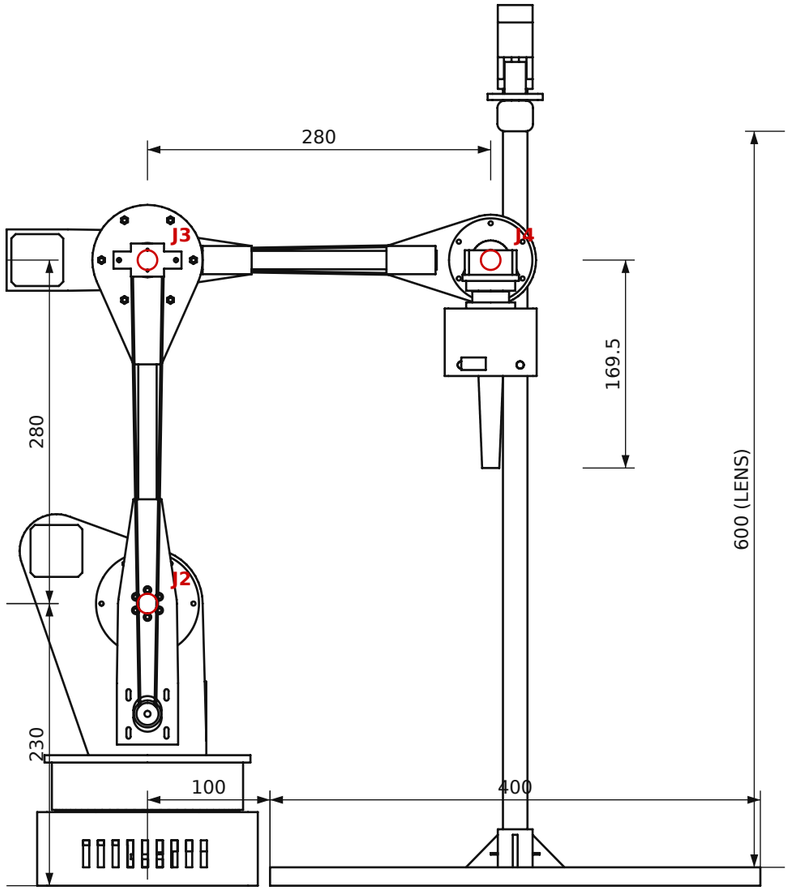
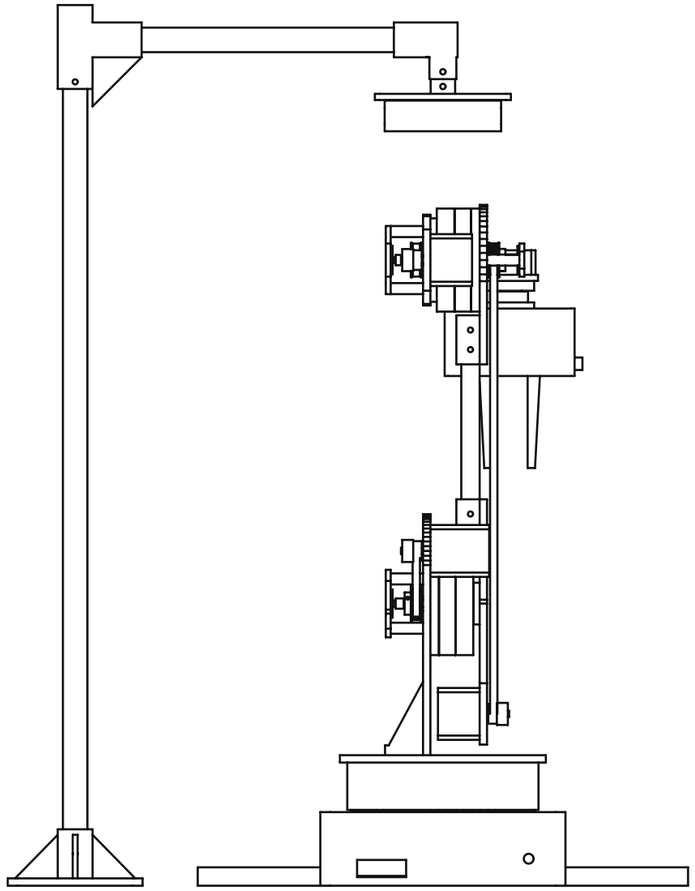
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

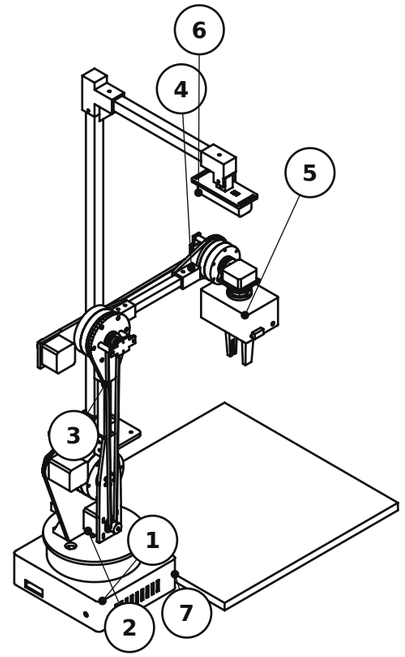


SIDE VIEW (FROM +X)

J2 SHOULDER, J3 ELBOW, J4 WRIST PITCH (RED); J1 YAW ABOUT THE BASE AXIS, J5 ROLL ABOUT THE GRIPPER AXIS



FRONT (FROM -Y)



ISOMETRIC (NOT TO SCALE)

ITEM	PART	DESCRIPTION	QTY	TYPE
1	base	electronics box, J1 drive, ball race	1	sheet 4
2	turret	turntable, tower, J2 drive + motor	1	sheet 5
3	upper arm	tube, fittings, J3 drive + motor	1	sheet 6
4	forearm	tube, fittings, J4 drive + motor	1	sheet 7
5	wrist + gripper	J5 roll, parallel gripper	1	sheet 8
6	camera pole	pole, boom, webcam plate	1	sheet 9
7	chessboard	40 x 40, not supplied	1	-

POSE SHOWN: J1 0°, J2 90°, J3 -90°, J4 -90° (UPPER ARM VERTICAL, FOREARM LEVEL).

CHESS ARM - GENERAL ARRANGEMENT

Whole build

PART / DWG No. GA-001

MATERIAL

SCALE	QTY	UNITS
1:5	-	mm

CHESS ROBOT ARM

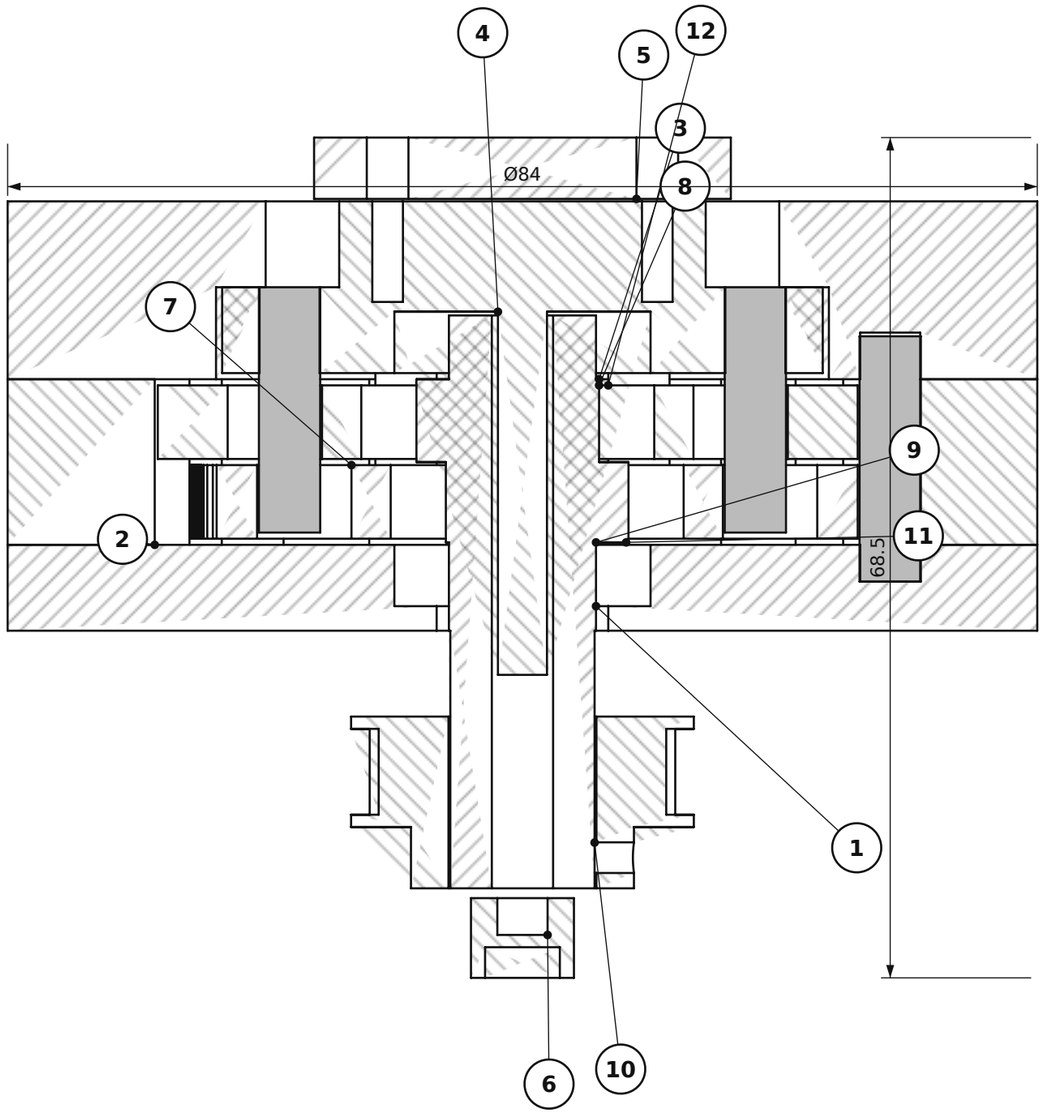
SHEET 1 / 51

2026-11-12

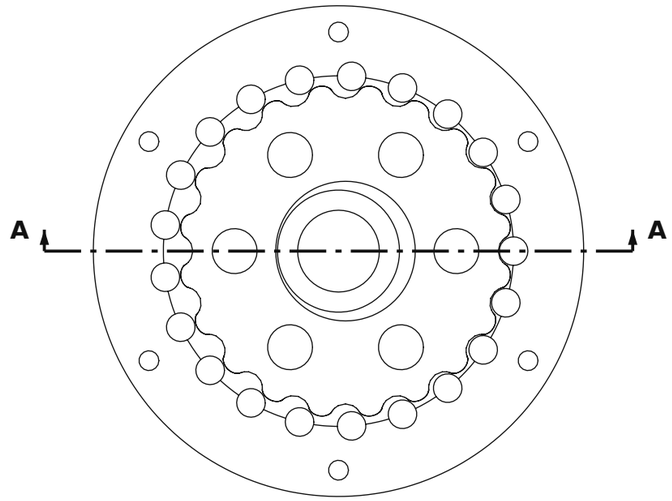
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

ITEM	PART	DESCRIPTION	QTY	TYPE
1	back cover	journal bearing 6801	1	printed
2	housing ring	21 ring pin pockets	1	printed
3	output cap	output bearing 6806 seat	1	printed
4	output flange	pin carrier + sensor rod	1	printed
5	output clamp	clamps output bearing	1	printed
6	magnet cap	6 x 2.5 magnet	1	printed
7	cycloid disc a	20 lobes, e = 1.2	1	printed
8	cycloid disc b	as A, 180° phase	1	printed
9	input shaft	2 eccentrics + 6802 x2	1	printed
10	input pulley	GT2 40T	1	printed
11	ring pins	Ø5 x 20 dowel x21	21	bought
12	output pins	Ø5 x 20 dowel x6	6	bought



SECTION A-A (2:1)



VIEW ON THE COVER (DISC SHOWN)

KEY DATA	
Ratio	20:1 (x2 belt = 40:1)
Pin circle	Ø60
Eccentricity	1.2 mm
Disc thickness	2 x 6
Bearings	6802 x2, 6801 x2, 6806
Stack bolts	6 x M3 on Ø75

CYCLOIDAL DRIVE J1-J3 (20:1)

Assembly, section

PART / DWG No. SA-010

MATERIAL

SCALE	QTY	UNITS
2:1	**3**	**mm**

CHESS ROBOT ARM

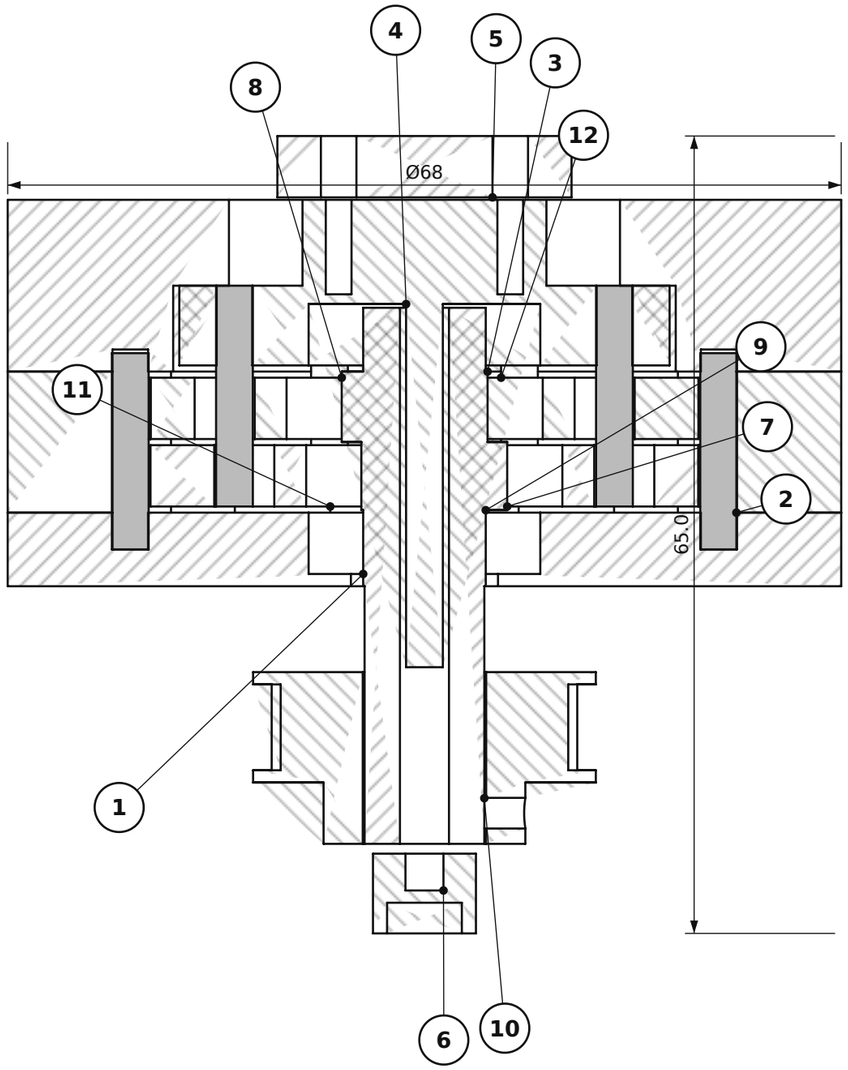
SHEET 2 / 51

2026-11-12

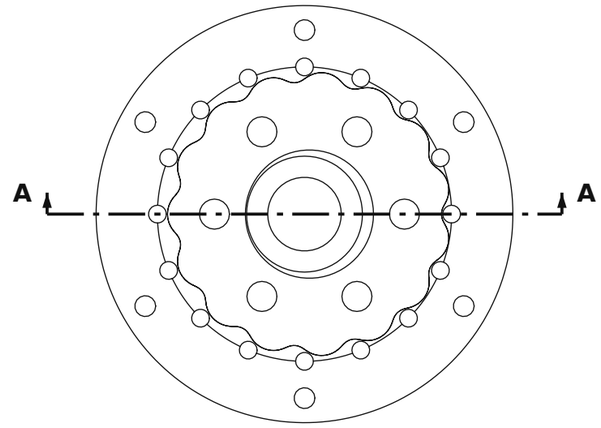
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

ITEM	PART	DESCRIPTION	QTY	TYPE
1	back cover	journal bearing 6800	1	printed
2	housing ring	16 ring pin pockets	1	printed
3	output cap	output bearing 6804 seat	1	printed
4	output flange	pin carrier + sensor rod	1	printed
5	output clamp	clamps output bearing	1	printed
6	magnet cap	6 x 2.5 magnet	1	printed
7	cycloid disc a	15 lobes, e = 0.8	1	printed
8	cycloid disc b	as A, 180° phase	1	printed
9	input shaft	2 eccentrics + 6801 x2	1	printed
10	input pulley	GT2 40T	1	printed
11	ring pins	Ø3 x 16 dowel x16	16	bought
12	output pins	Ø3 x 18 dowel x6	6	bought



SECTION A-A (2:1)



VIEW ON THE COVER (DISC SHOWN)

KEY DATA	
Ratio	15:1 (x2 belt = 30:1)
Pin circle	Ø48
Eccentricity	0.8 mm
Disc thickness	2 x 5
Bearings	6801 x2, 6800 x2, 6804
Stack bolts	6 x M3 on Ø60

CYCLOIDAL DRIVE J4 (15:1)

Assembly, section

PART / DWG No. SA-011

MATERIAL

SCALE	QTY	UNITS
2:1	**1**	**mm**

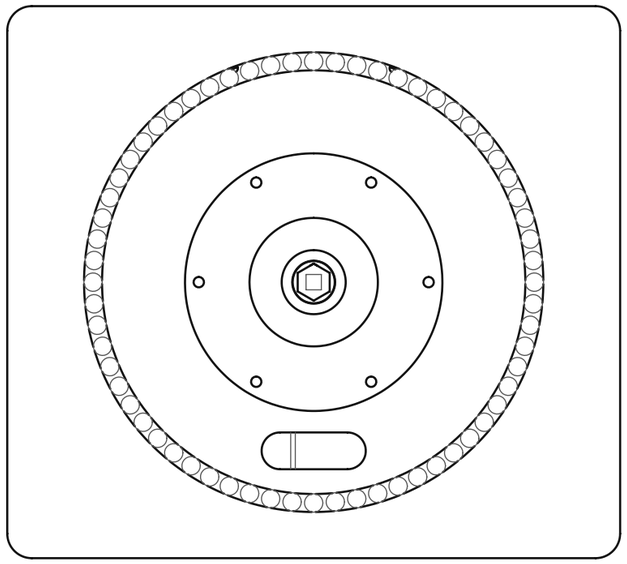
CHESS ROBOT ARM

SHEET 3 / 51

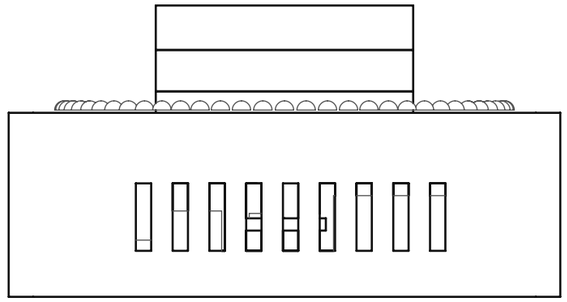
2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

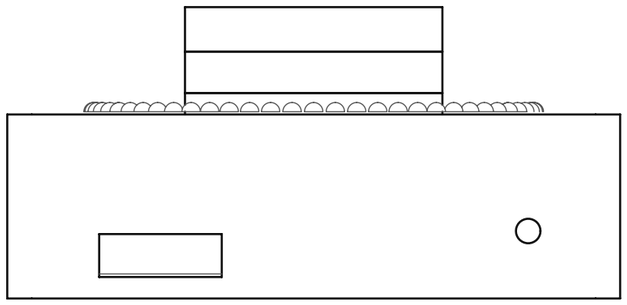


TOP

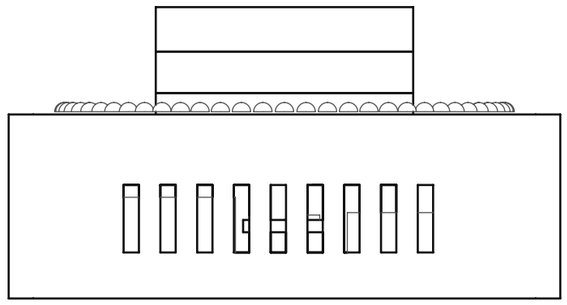


LEFT

ITEM	PART	DESCRIPTION	QTY	TYPE
1	base shell	printed PETG	1	printed
2	base floor	printed PETG	1	printed
3	j1 motor shelf	printed PETG	1	printed
4	J1 sensor bridge	printed PETG	1	printed
5	J1 back cover	printed PETG	1	printed
6	J1 housing ring	printed PETG	1	printed
7	J1 output cap	printed PETG	1	printed
8	J1 input pulley	printed PETG	1	printed
9	J1 motor	stepper motor (see BOM)	1	bought
10	J1 motor pulley	GT2 20T pulley, 5 mm bore	1	bought
11	J1 belt	GT2 6 mm closed belt	1	bought
12	J1 as5600	AS5600 board + magnet	1	bought
13	race balls	Ø6 steel balls	1	bought
14	computer	Raspberry Pi 4	1	bought
15	esp32 board	ESP32 + driver perfboard	1	bought
16	buck	24 V -> 5 V buck	1	bought
17	oled	OLED 1.3"	1	bought



FRONT



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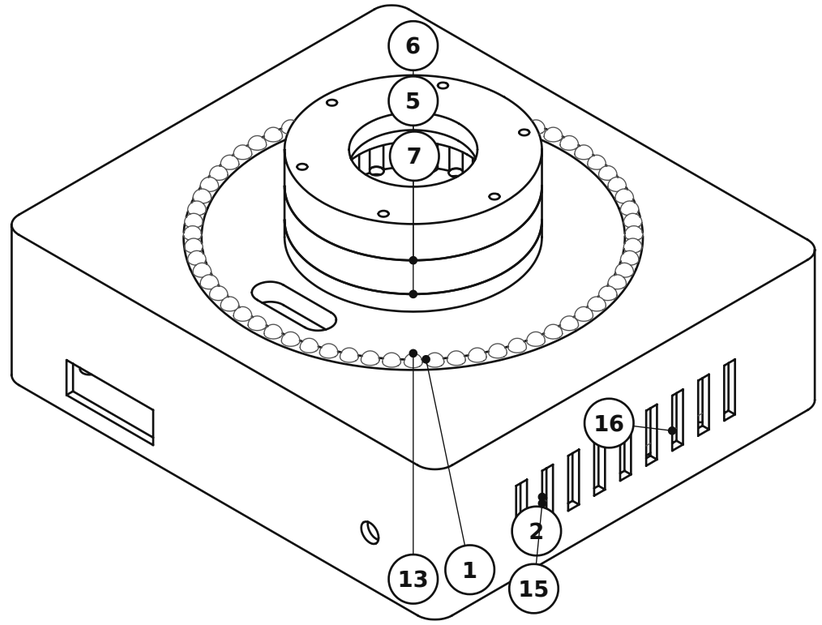


ISOMETRIC (NOT TO SCALE)

BASE

Assembly

PART / DWG No. SA-001

MATERIAL

SCALE	QTY	UNITS
1:2	**-**	**mm**

CHESS ROBOT ARM

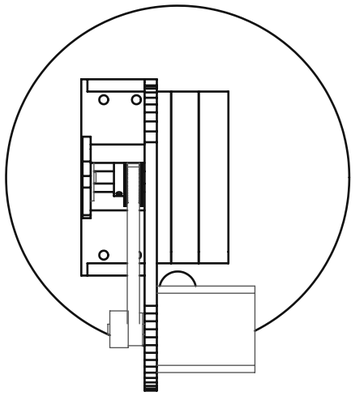
SHEET 4 / 51

2026-11-12

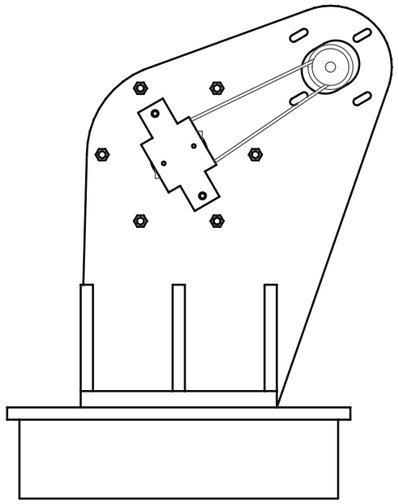
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

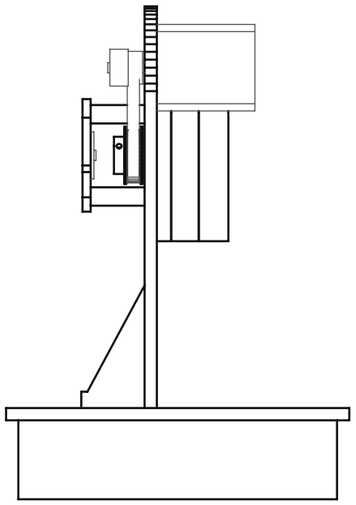
ITEM	PART	DESCRIPTION	QTY	TYPE
1	turntable	printed PETG	1	printed
2	tower	printed PETG	1	printed
3	J2 sensor bridge	printed PETG	1	printed
4	J1 output clamp	printed PETG	1	printed
5	J2 back cover	printed PETG	1	printed
6	J2 housing ring	printed PETG	1	printed
7	J2 output cap	printed PETG	1	printed
8	J2 input pulley	printed PETG	1	printed
9	J2 motor	stepper motor (see BOM)	1	bought
10	J2 motor pulley	GT2 20T pulley, 5 mm bore	1	bought
11	J2 belt	GT2 6 mm closed belt	1	bought
12	J2 as5600	AS5600 board + magnet	1	bought



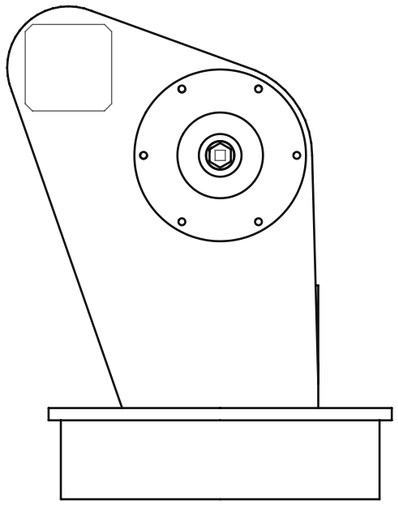
TOP



LEFT



FRONT



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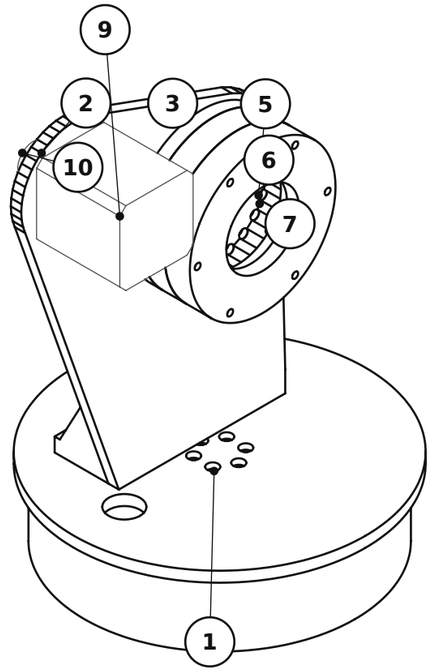


ISOMETRIC (NOT TO SCALE)

TURRET (TURNTABLE, TOWER, J2)

Assembly

PART / DWG No. SA-002

MATERIAL

SCALE	QTY	UNITS
1:3	**-**	**mm**

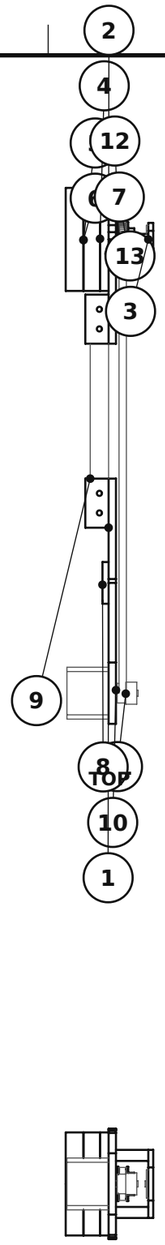
CHESS ROBOT ARM

SHEET 5 / 51

2026-11-12

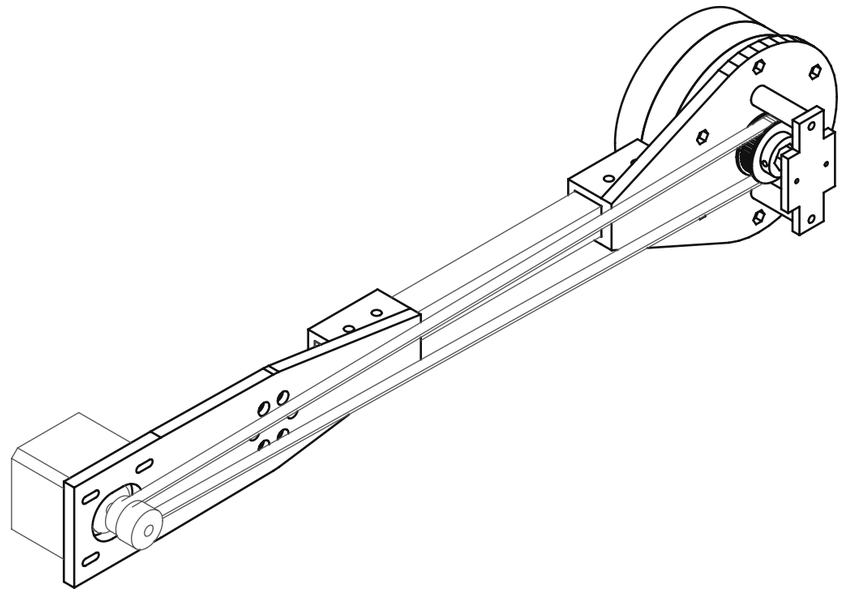
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



FRONT

ITEM	PART	DESCRIPTION	QTY	TYPE
1	upper proximal	printed PETG	1	printed
2	upper distal	printed PETG	1	printed
3	J3 sensor bridge	printed PETG	1	printed
4	J3 back cover	printed PETG	1	printed
5	J3 housing ring	printed PETG	1	printed
6	J3 output cap	printed PETG	1	printed
7	J3 input pulley	printed PETG	1	printed
8	J2 output clamp	printed PETG	1	printed
9	upper tube	aluminium square tube	1	bought
10	J3 motor	stepper motor (see BOM)	1	bought
11	J3 motor pulley	GT2 20T pulley, 5 mm bore	1	bought
12	J3 belt	GT2 6 mm closed belt	1	bought
13	J3 as5600	AS5600 board + magnet	1	bought



ISOMETRIC (NOT TO SCALE)

UPPER ARM

Assembly

PART / DWG No. SA-003

MATERIAL

SCALE	QTY	UNITS
1:5	**-**	**mm**

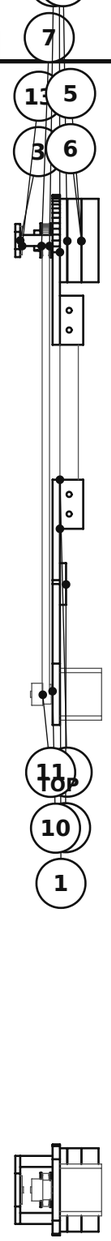
CHESS ROBOT ARM

SHEET 6 / 51

2026-11-12

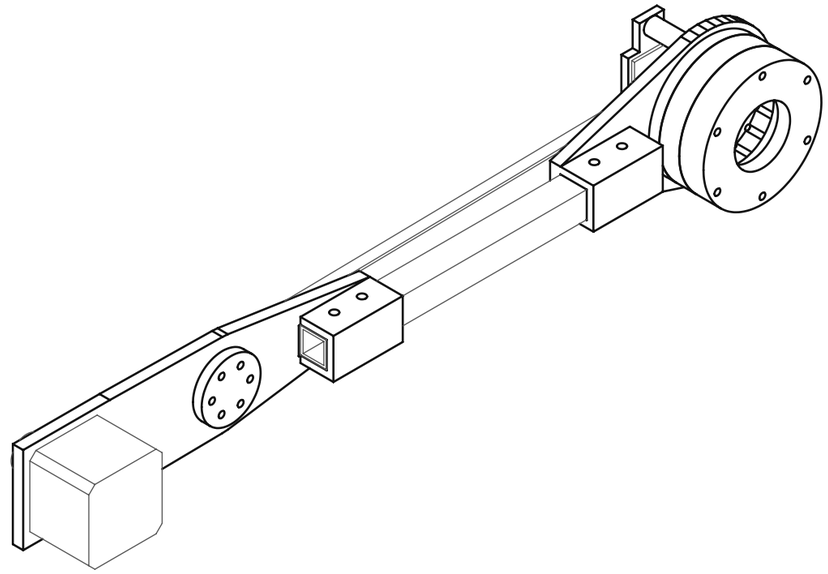
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



FRONT

ITEM	PART	DESCRIPTION	QTY	TYPE
1	fore proximal	printed PETG	1	printed
2	fore distal	printed PETG	1	printed
3	J4 sensor bridge	printed PETG	1	printed
4	J4 back cover	printed PETG	1	printed
5	J4 housing ring	printed PETG	1	printed
6	J4 output cap	printed PETG	1	printed
7	J4 input pulley	printed PETG	1	printed
8	J3 output clamp	printed PETG	1	printed
9	fore tube	aluminium square tube	1	bought
10	J4 motor	stepper motor (see BOM)	1	bought
11	J4 motor pulley	GT2 20T pulley, 5 mm bore	1	bought
12	J4 belt	GT2 6 mm closed belt	1	bought
13	J4 as5600	AS5600 board + magnet	1	bought



ISOMETRIC (NOT TO SCALE)

FOREARM

Assembly

PART / DWG No. SA-004

MATERIAL

SCALE	QTY	UNITS
1:5	**-**	**mm**

CHESS ROBOT ARM

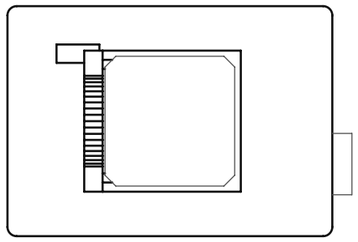
SHEET 7 / 51

2026-11-12

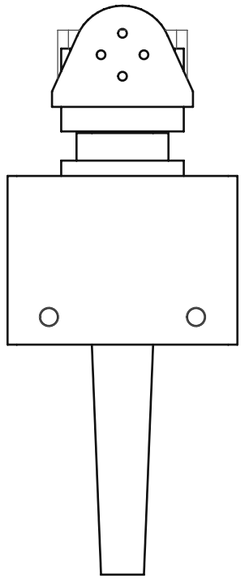
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

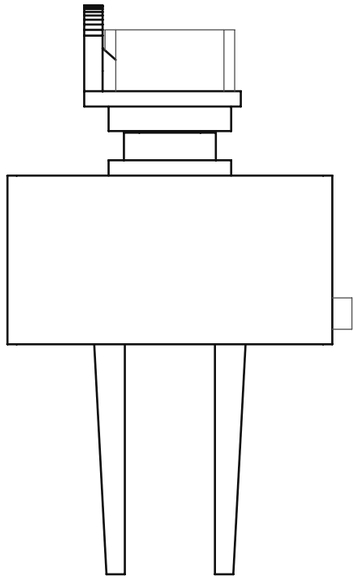
ITEM	PART	DESCRIPTION	QTY	TYPE
1	wrist bracket	printed PETG	1	printed
2	roll hub	printed PETG	1	printed
3	gripper frame	printed PETG	1	printed
4	pinion	printed PETG	1	printed
5	jaw a	printed PETG	1	printed
6	jaw b	printed PETG	1	printed
7	roll motor	stepper motor (see BOM)	1	bought
8	roll bearing	6804 bearing	1	bought
9	gripper motor	stepper motor (see BOM)	1	bought
10	gripper rods	Ø6 steel rod x2	1	bought
11	lm6uu	LM6UU bushing x4	1	bought
12	home switch	micro switch	1	bought



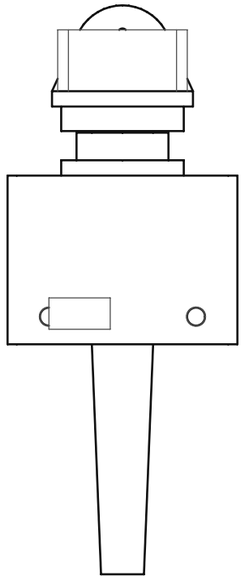
TOP



LEFT



FRONT



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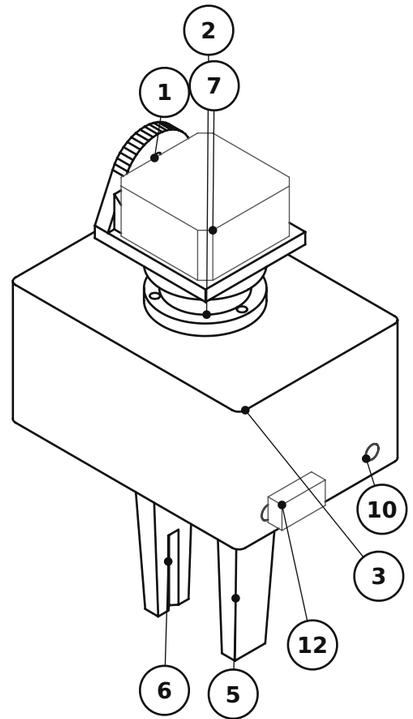


ISOMETRIC (NOT TO SCALE)

WRIST + GRIPPER

Assembly

PART / DWG No. SA-005

MATERIAL

SCALE	QTY	UNITS
1:2	**-**	**mm**

CHESS ROBOT ARM

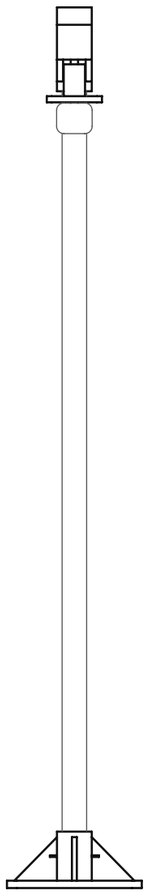
SHEET 8 / 51

2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

ITEM	PART	DESCRIPTION	QTY	TYPE
1	camera foot	printed PETG	1	printed
2	camera tee	printed PETG	1	printed
3	camera mount	printed PETG	1	printed
4	camera plate	printed PETG	1	printed
5	pole tube	aluminium square tube	1	bought
6	boom tube	aluminium square tube	1	bought
7	webcam	webcam	1	bought



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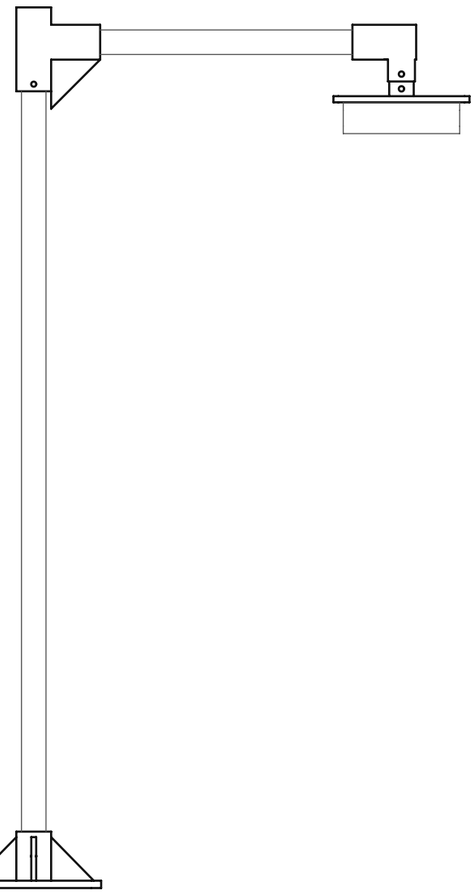
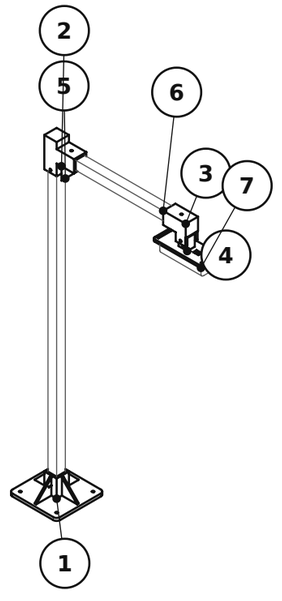


FRONT



ISOMETRIC (NOT TO SCALE)

CAMERA POLE

Assembly

PART / DWG No. SA-006

MATERIAL

SCALE	QTY	UNITS
1:5	**-**	**mm**

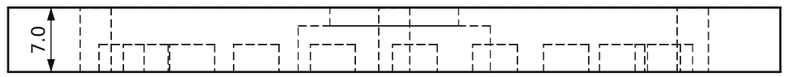
CHESS ROBOT ARM

SHEET 9 / 51

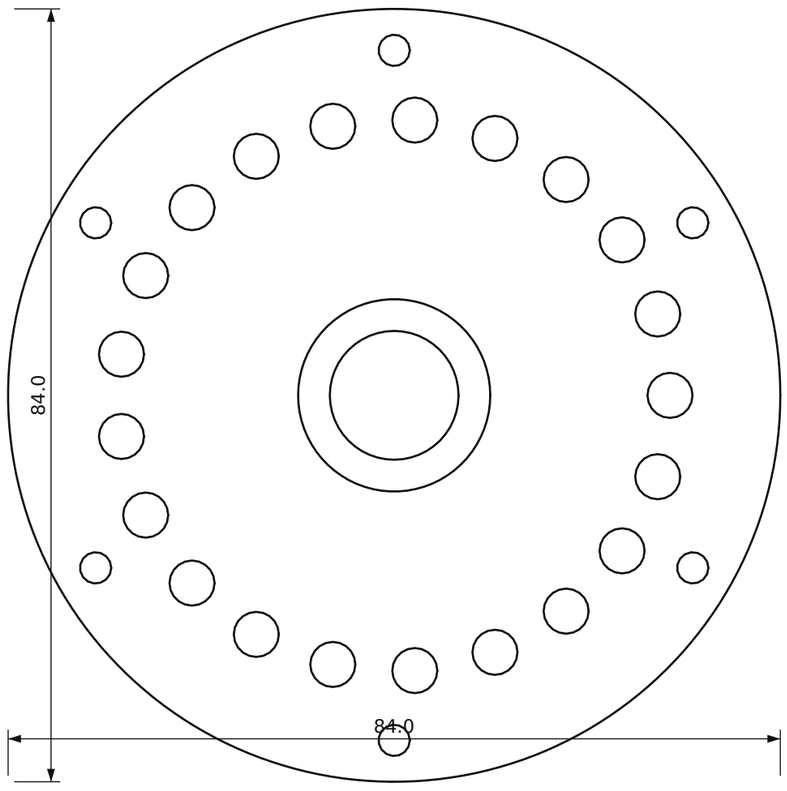
2026-11-12

ALL DIMENSIONS IN MM

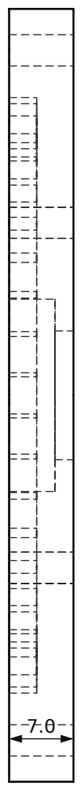
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



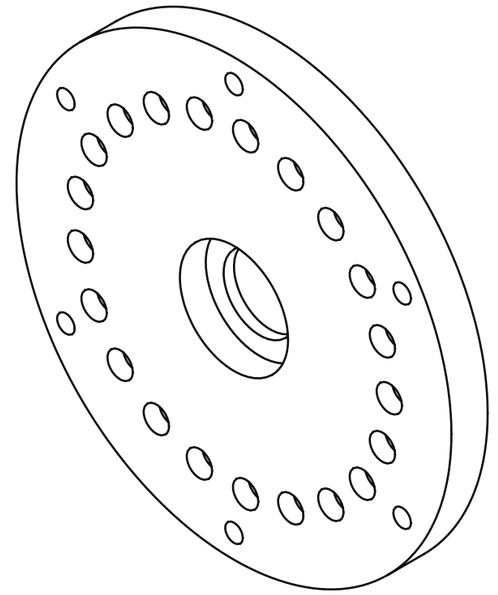
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø14	Y	1
Ø3.4	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

BACK COVER

back_cover (J1, J2, J3)

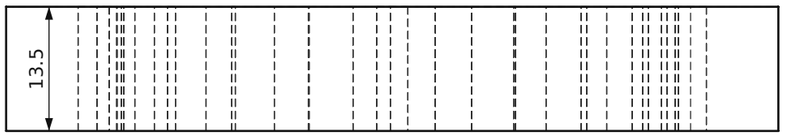
PART / DWG No. P-010
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**3**	**mm**

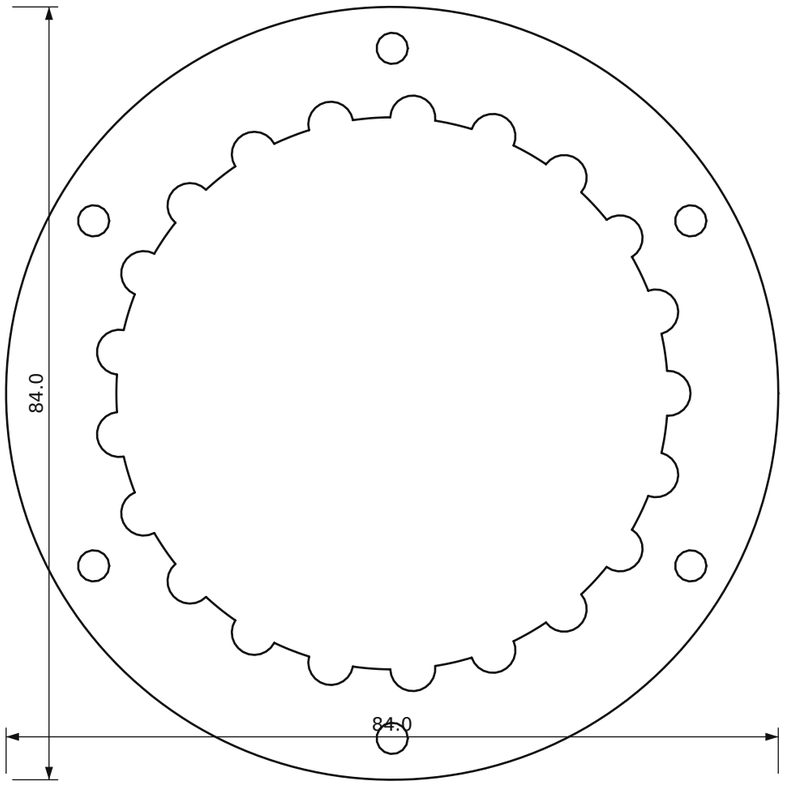
CHESS ROBOT ARM
SHEET 10 / 51
2026-11-12

ALL DIMENSIONS IN MM

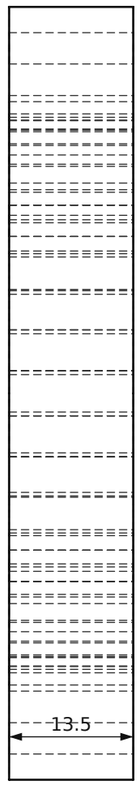
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



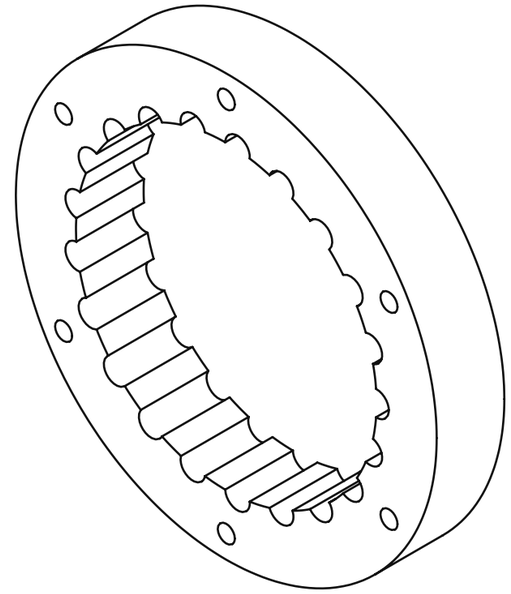
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø60	Y	1
Ø4.9	Y	21
Ø3.4	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

HOUSING RING

housing_ring (J1, J2, J3)

PART / DWG No. P-011
MATERIAL PETG

SCALE	QTY	UNITS
3:2	3	mm

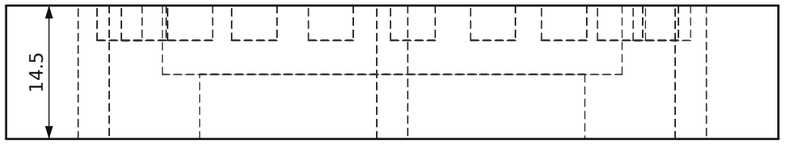
CHESS ROBOT ARM

SHEET 11 / 51

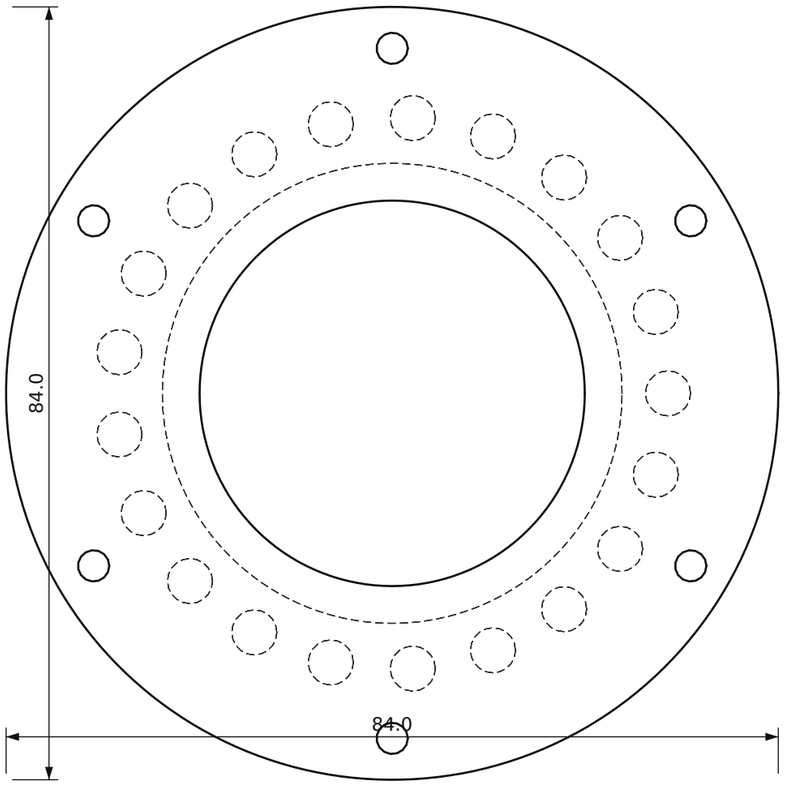
2026-11-12

ALL DIMENSIONS IN MM

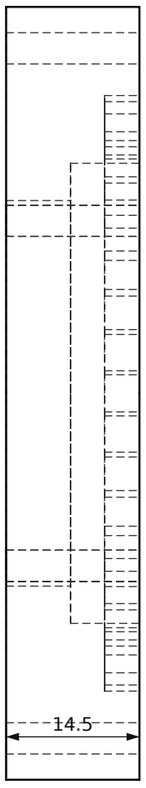
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



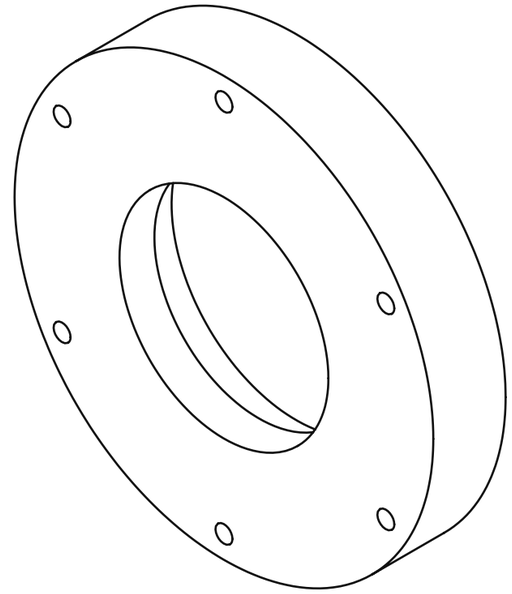
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø50	Y	1
Ø41.9	Y	1
Ø4.9	Y	21
Ø3.4	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT CAP

output_cap (J1, J2, J3)

PART / DWG No. P-012
MATERIAL PETG

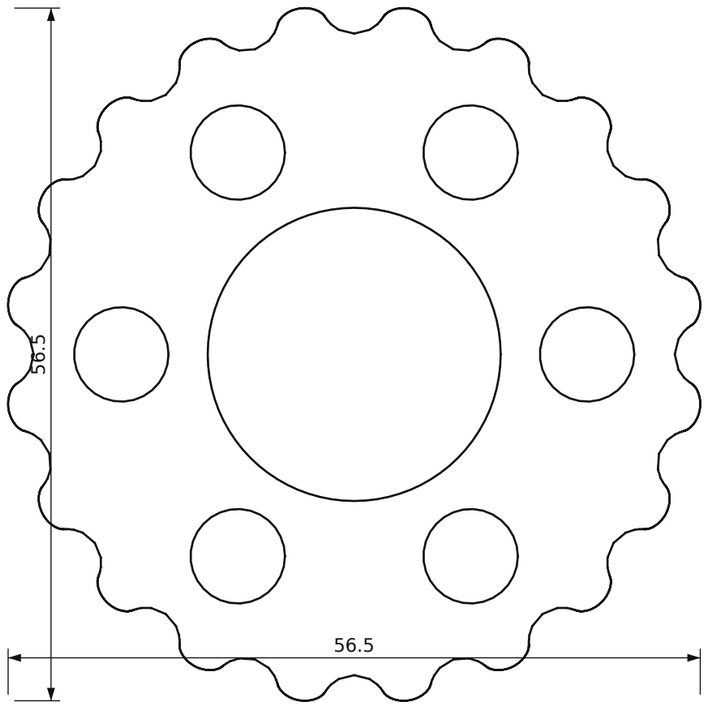
SCALE	QTY	UNITS
3:2	**3**	**mm**

CHESS ROBOT ARM
SHEET 12 / 51
2026-11-12

ALL DIMENSIONS IN MM

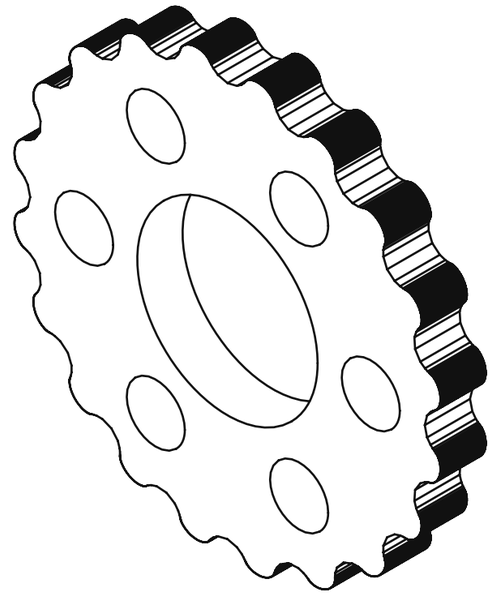
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

TOP



FRONT

RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø23.9	Y	1
Ø7.7	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CYCLOID DISC A

cycloid_disc_a (J1, J2, J3)

PART / DWG No. P-013
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**3**	**mm**

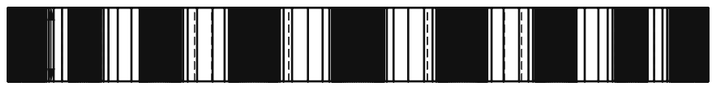
CHESS ROBOT ARM

SHEET 13 / 51

2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



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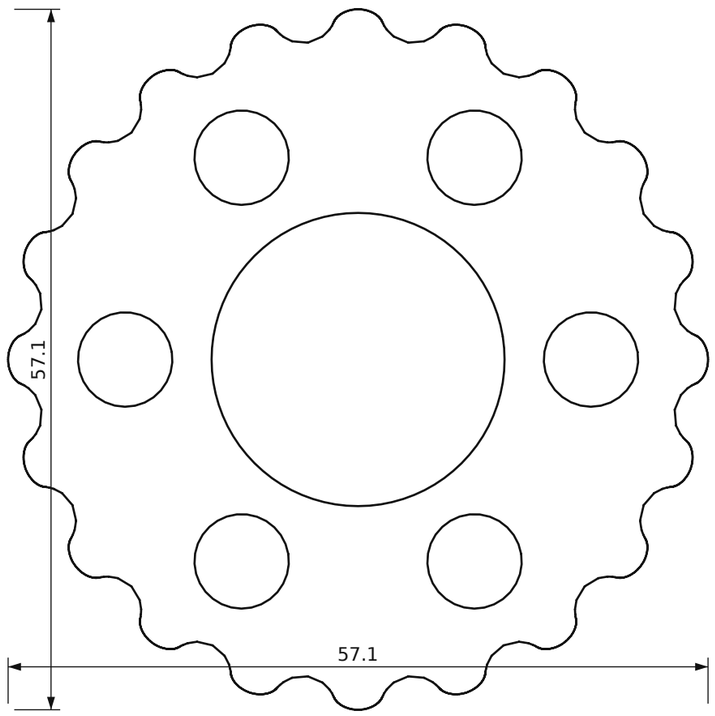
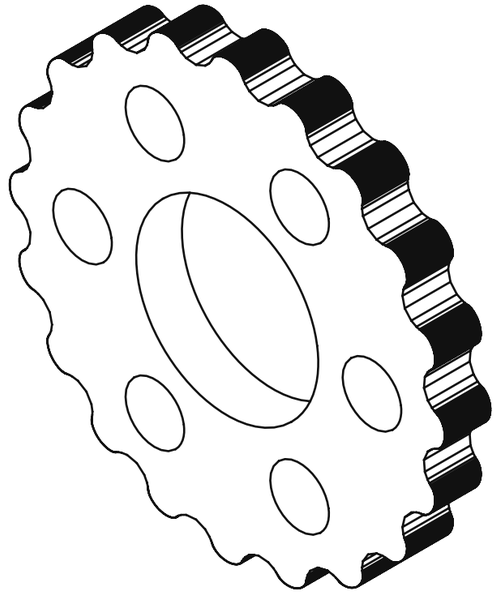


FRONT

RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø23.9	Y	1
Ø7.7	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CYCLOID DISC B

cycloid_disc_b (J1, J2, J3)

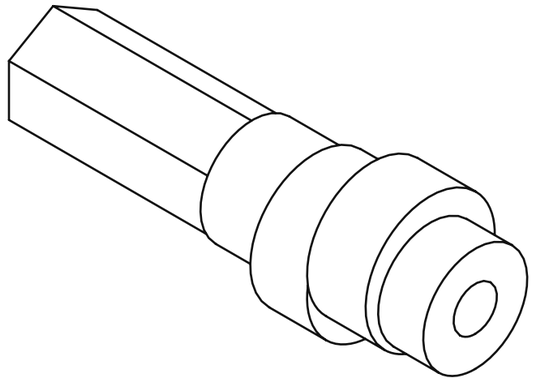
PART / DWG No. P-014
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**3**	**mm**

CHESS ROBOT ARM
SHEET 14 / 51
2026-11-12

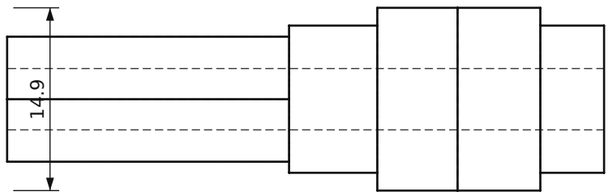
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



ISOMETRIC

HOLE	AXIS	QTY
Ø14.9	X	1
Ø12	X	1
Ø5	X	1



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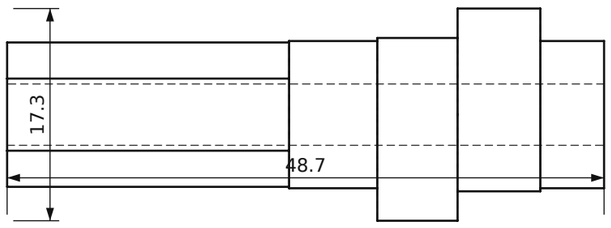
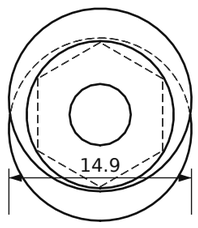


FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

INPUT SHAFT

input_shaft (J1)

PART / DWG No. P-015
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

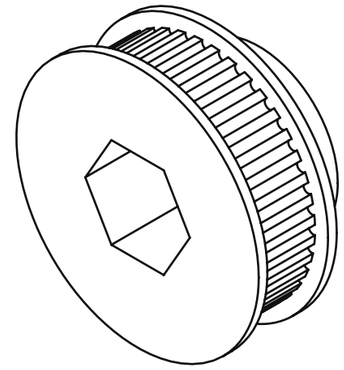
CHESS ROBOT ARM

SHEET 15 / 51

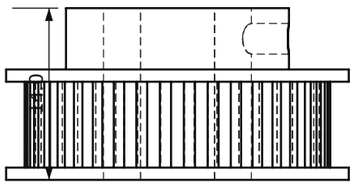
2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



ISOMETRIC



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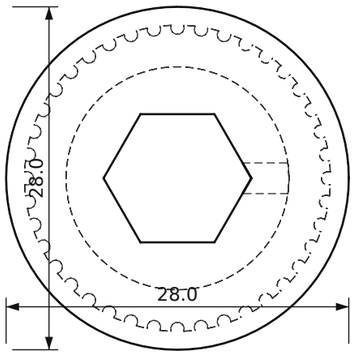
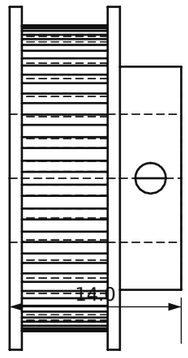


FRONT



RIGHT

HOLE	AXIS	QTY
Ø27.96	Y	1
Ø2.5	X	1

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

INPUT PULLEY

input_pulley (J1, J2, J3)

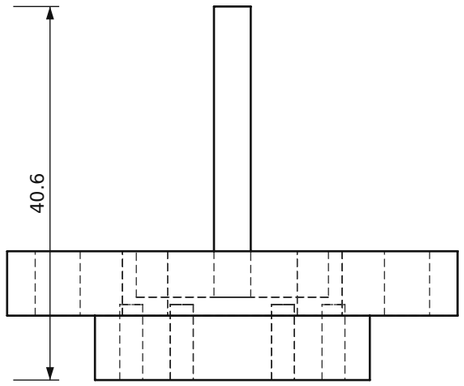
PART / DWG No. P-016
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**3**	**mm**

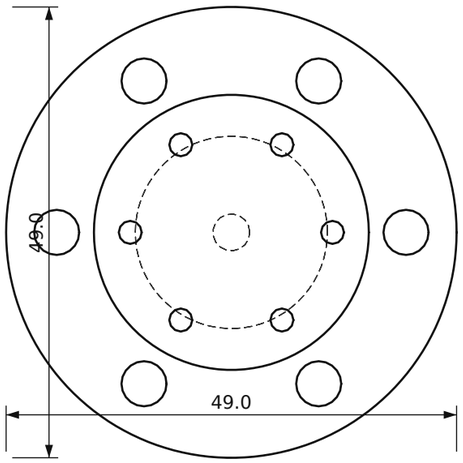
CHESS ROBOT ARM
SHEET 16 / 51
2026-11-12

ALL DIMENSIONS IN MM

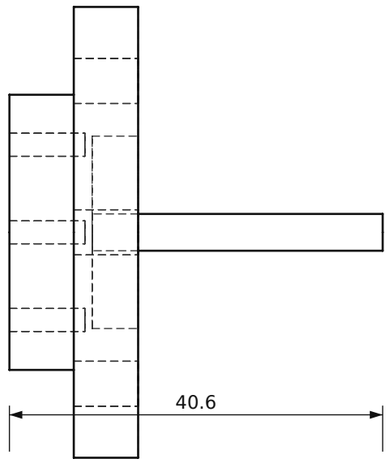
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



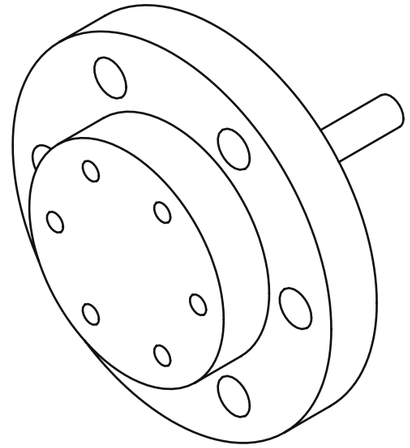
TOP



FRONT



RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø49	Y	1
Ø20.9	Y	1
Ø4.9	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT FLANGE

output_flange (J1)

PART / DWG No. P-017
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

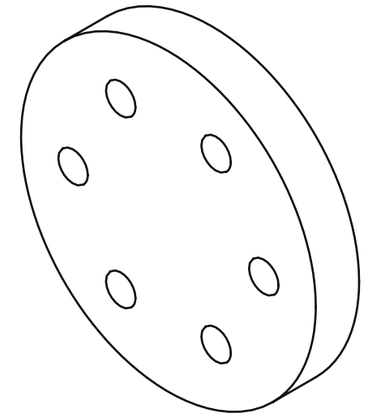
CHESS ROBOT ARM

SHEET 17 / 51

2026-11-12

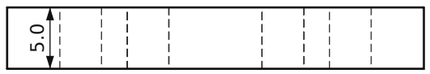
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

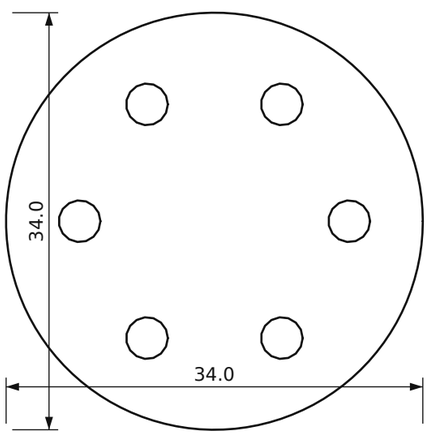


ISOMETRIC

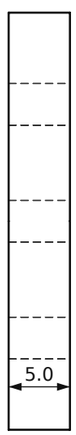
HOLE	AXIS	QTY
Ø34	Y	1
Ø3.4	Y	6



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT CLAMP

output_clamp (J1, J2, J3)

PART / DWG No. P-018
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**3**	**mm**

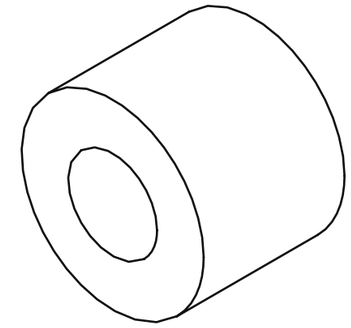
CHESS ROBOT ARM

SHEET 18 / 51

2026-11-12

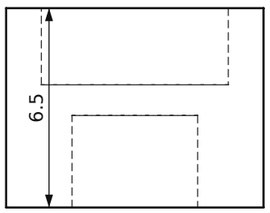
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

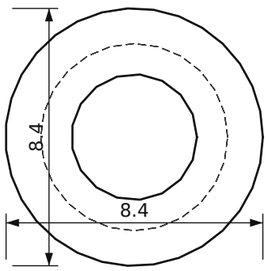


ISOMETRIC

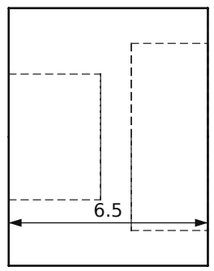
HOLE	AXIS	QTY
Ø8.4	Y	1
Ø6.1	Y	1



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

MAGNET CAP

magnet_cap (J1, J2, J3)

PART / DWG No. P-019
MATERIAL PETG

SCALE	QTY	UNITS
5:1	**3**	**mm**

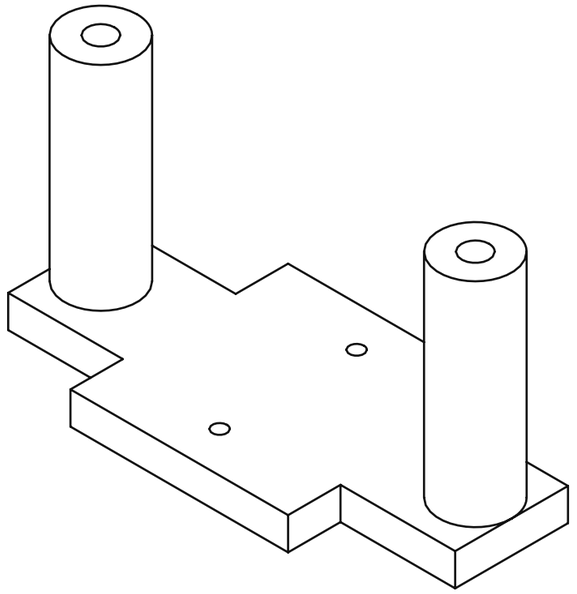
CHESS ROBOT ARM

SHEET 19 / 51

2026-11-12

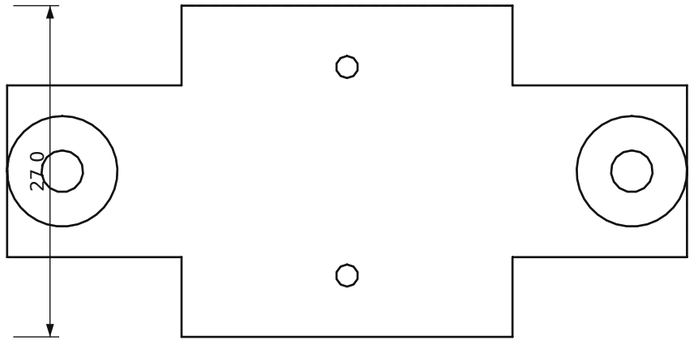
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

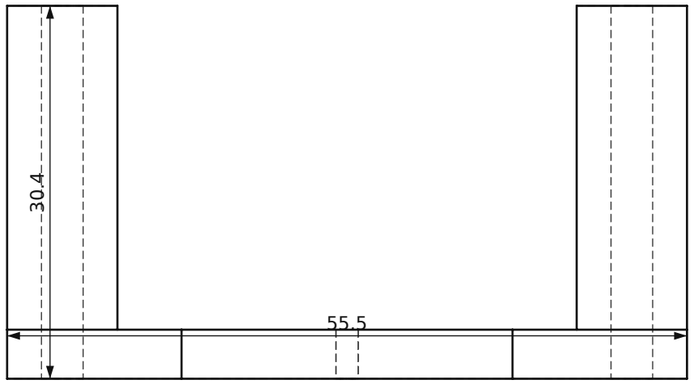


ISOMETRIC (NOT TO SCALE)

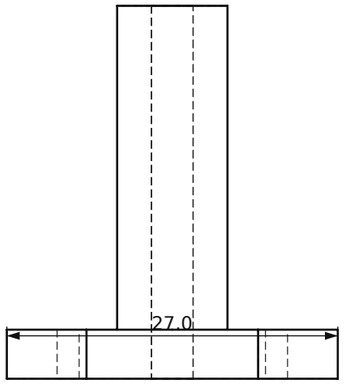
HOLE	AXIS	QTY
Ø9	Z	2
Ø3.4	Z	2
Ø1.8	Z	2



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

SENSOR BRIDGE

sensor_bridge (J1, J2, J3, J4)

PART / DWG No. P-020
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**4**	**mm**

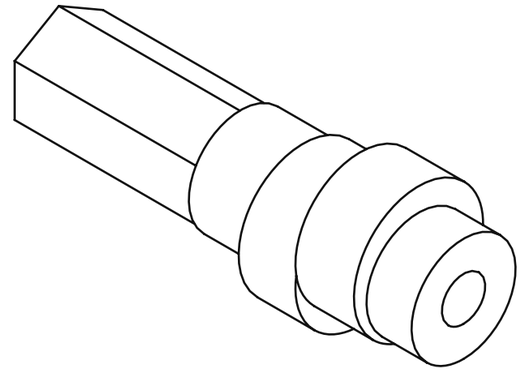
CHESS ROBOT ARM

SHEET 20 / 51

2026-11-12

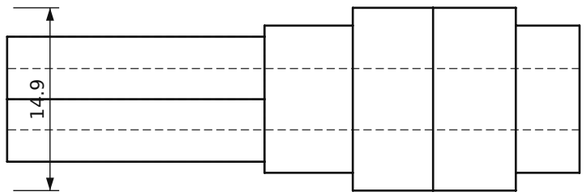
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

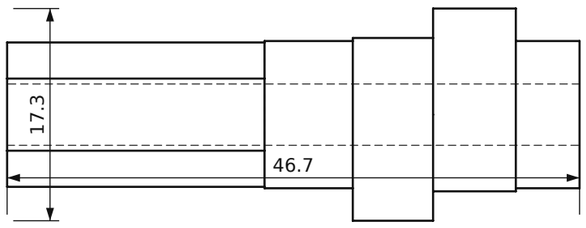


ISOMETRIC

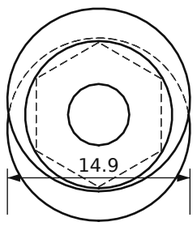
HOLE	AXIS	QTY
Ø14.9	X	1
Ø12	X	1
Ø5	X	1



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

INPUT SHAFT

input_shaft (J2, J3)

PART / DWG No. P-021
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**2**	**mm**

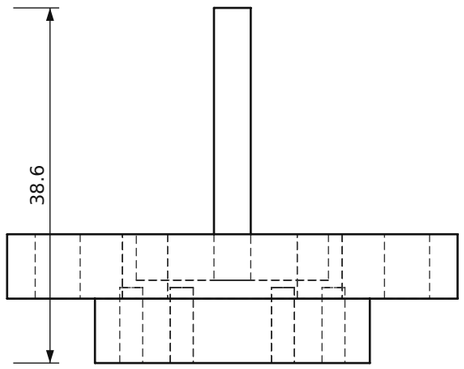
CHESS ROBOT ARM

SHEET 21 / 51

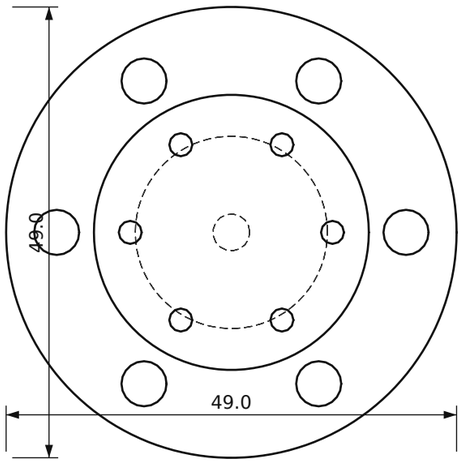
2026-11-12

ALL DIMENSIONS IN MM

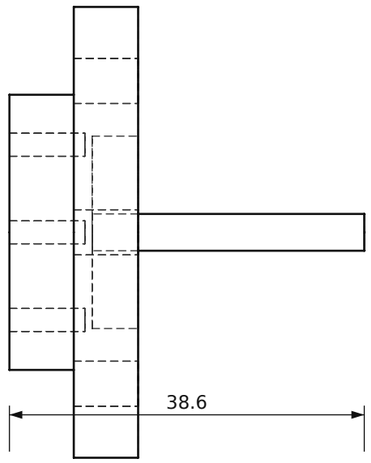
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



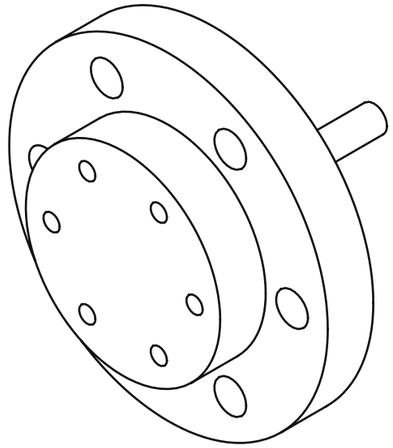
TOP



FRONT



RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø49	Y	1
Ø20.9	Y	1
Ø4.9	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT FLANGE

output_flange (J2, J3)

PART / DWG No. P-022
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**2**	**mm**

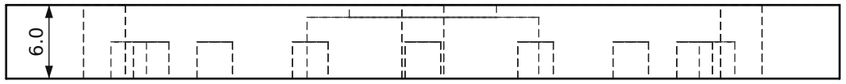
CHESS ROBOT ARM

SHEET 22 / 51

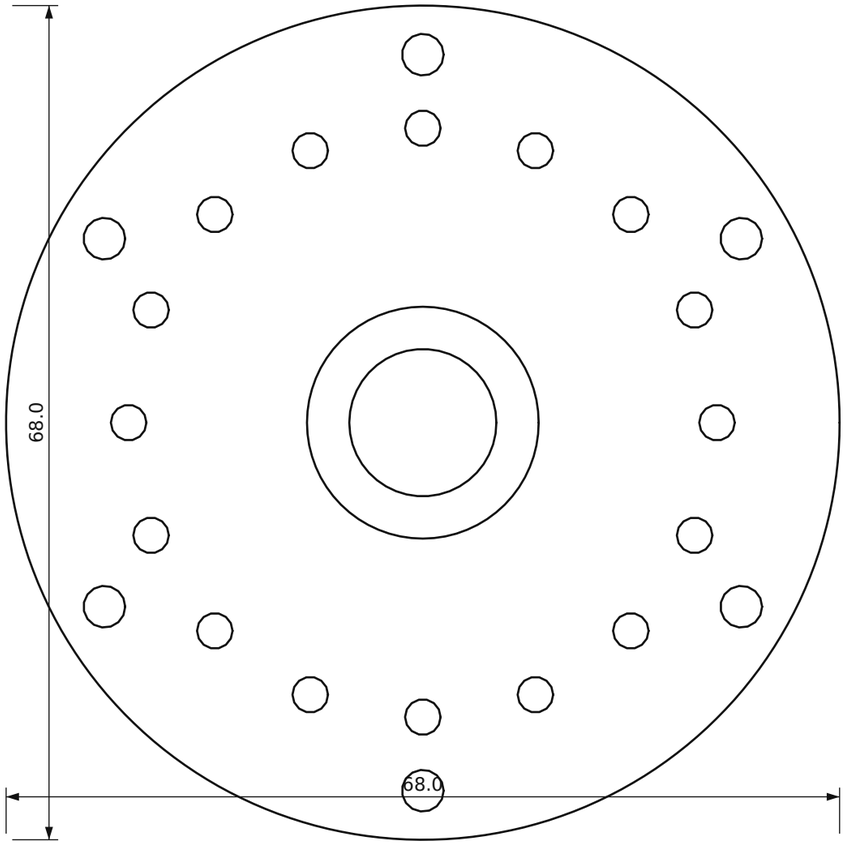
2026-11-12

ALL DIMENSIONS IN MM

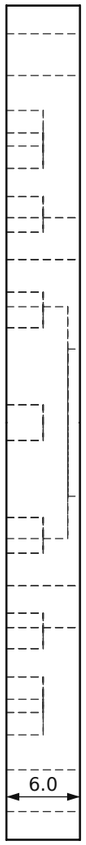
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



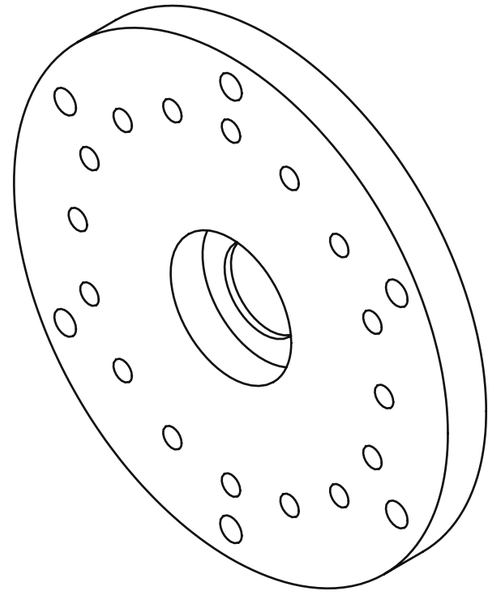
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø68	Y	1
Ø12	Y	1
Ø3.4	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

BACK COVER

back_cover (J4)

PART / DWG No. P-023
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

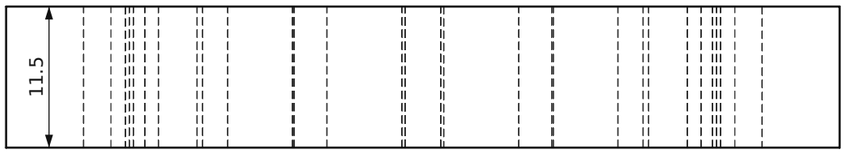
CHESS ROBOT ARM

SHEET 23 / 51

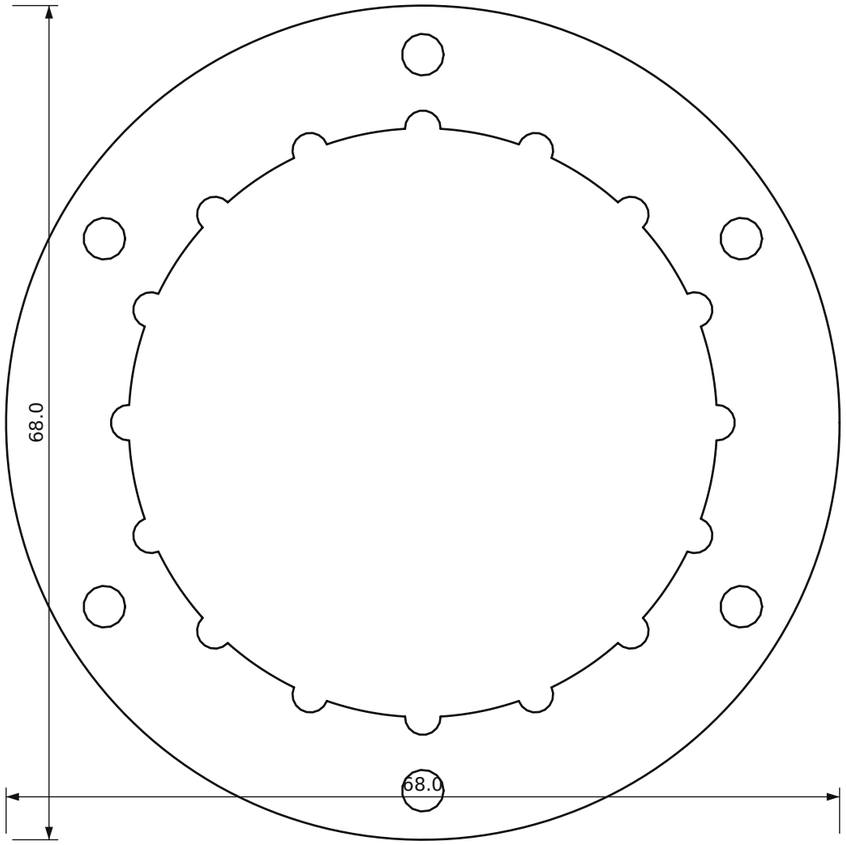
2026-11-12

ALL DIMENSIONS IN MM

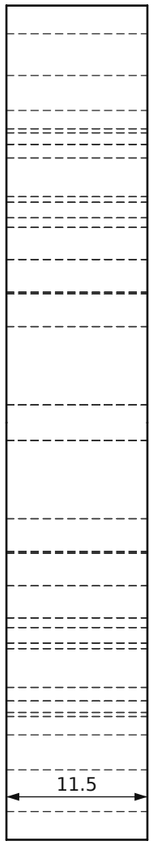
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



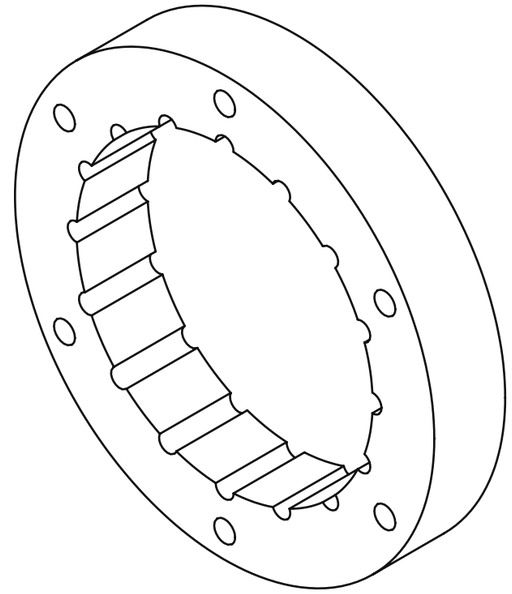
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø68	Y	1
Ø48	Y	1
Ø3.4	Y	6
Ø2.9	Y	16

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

HOUSING RING

housing_ring (J4)

PART / DWG No. P-024
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

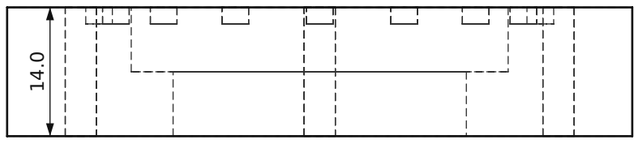
CHESS ROBOT ARM

SHEET 24 / 51

2026-11-12

ALL DIMENSIONS IN MM

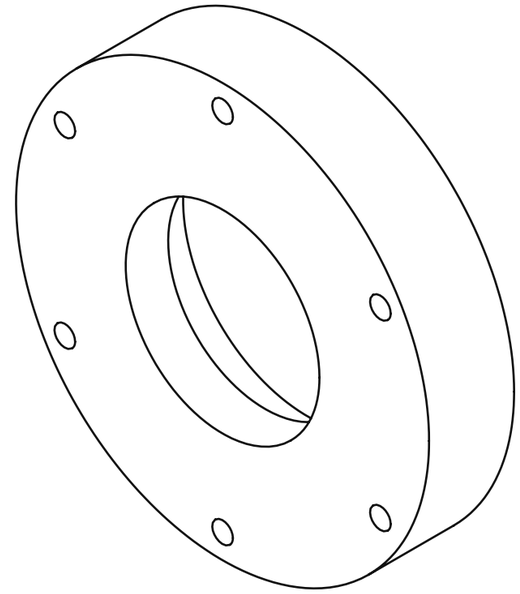
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



TOP

FRONT

RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø68	Y	1
Ø41	Y	1
Ø31.9	Y	1
Ø3.4	Y	6
Ø2.9	Y	16

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT CAP

output_cap (J4)

PART / DWG No. P-025
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

CHESS ROBOT ARM

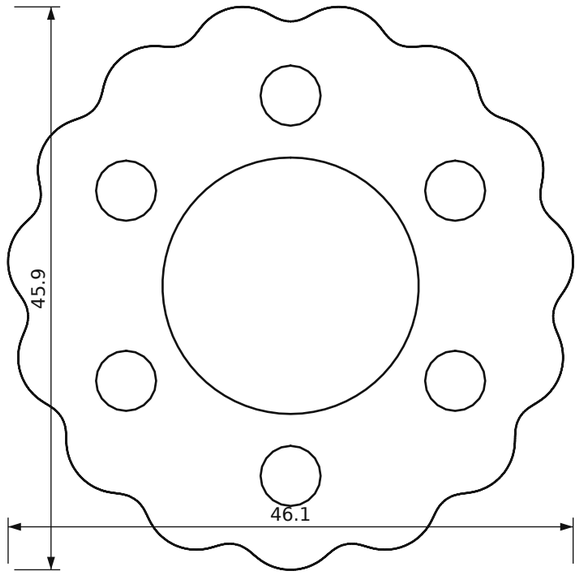
SHEET 25 / 51

2026-11-12

ALL DIMENSIONS IN MM

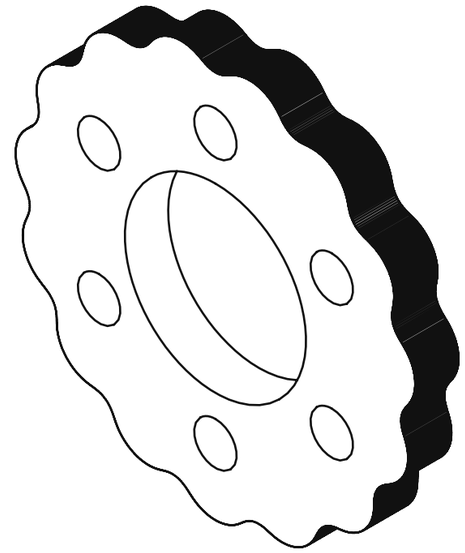
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

TOP



FRONT

RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø20.9	Y	1
Ø4.9	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CYCLOID DISC A

cycloid_disc_a (J4)

PART / DWG No. P-026
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

CHESS ROBOT ARM

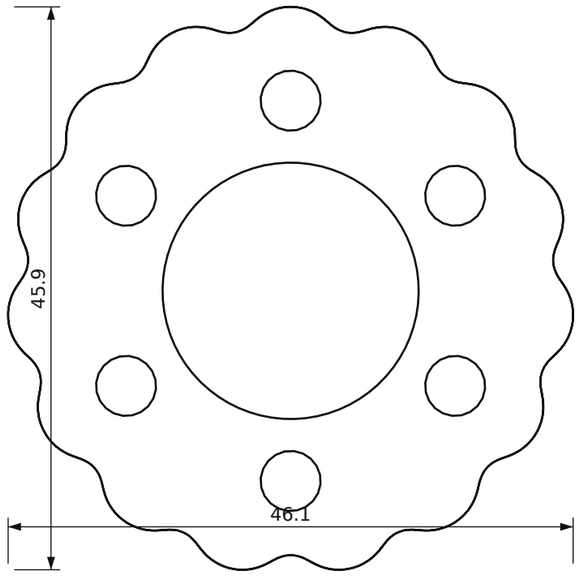
SHEET 26 / 51

2026-11-12

ALL DIMENSIONS IN MM

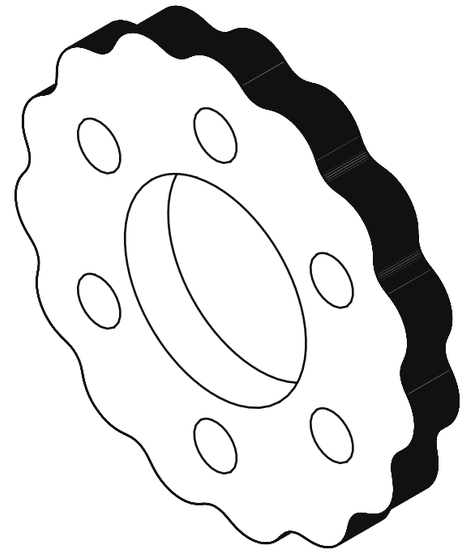
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

TOP



FRONT

RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø20.9	Y	1
Ø4.9	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CYCLOID DISC B

cycloid_disc_b (J4)

PART / DWG No. P-027
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

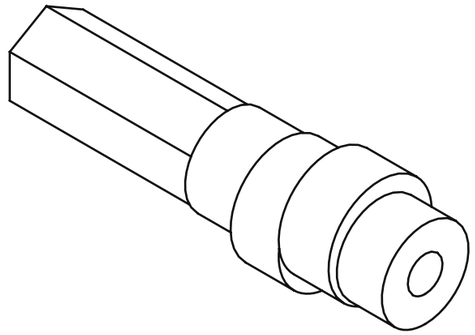
CHESS ROBOT ARM

SHEET 27 / 51

2026-11-12

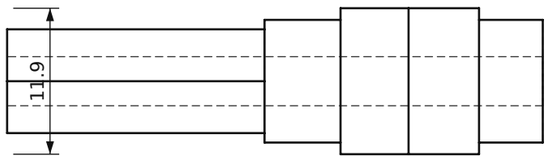
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

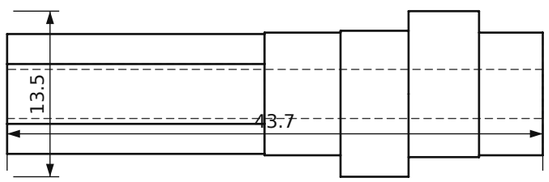


ISOMETRIC

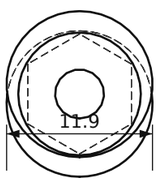
HOLE	AXIS	QTY
Ø11.9	X	1
Ø10	X	1
Ø4	X	1



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

INPUT SHAFT

input_shaft (J4)

PART / DWG No. P-028
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

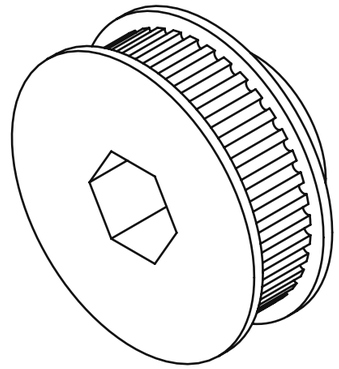
CHESS ROBOT ARM

SHEET 28 / 51

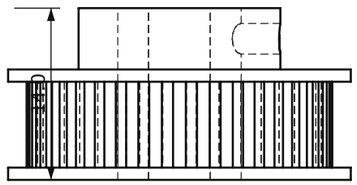
2026-11-12

ALL DIMENSIONS IN MM

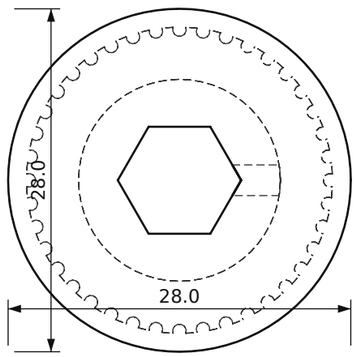
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



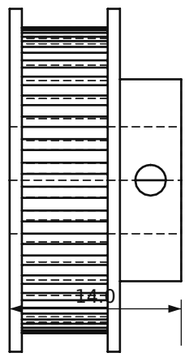
ISOMETRIC



TOP



FRONT



RIGHT

HOLE	AXIS	QTY
Ø27.96	Y	1
Ø2.5	X	1

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

INPUT PULLEY

input_pulley (J4)

PART / DWG No. P-029
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

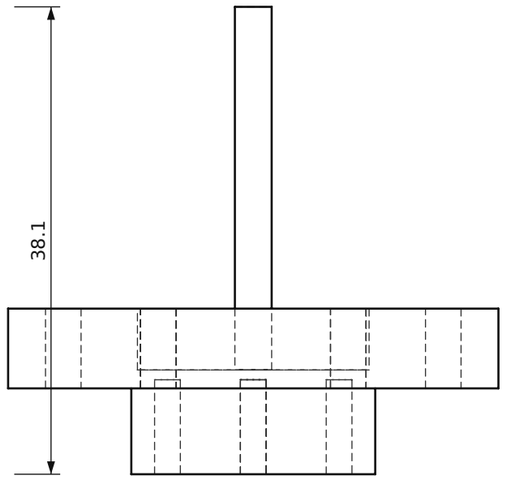
CHESS ROBOT ARM

SHEET 29 / 51

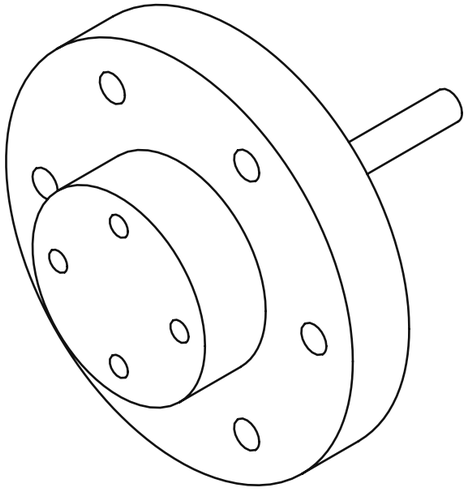
2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

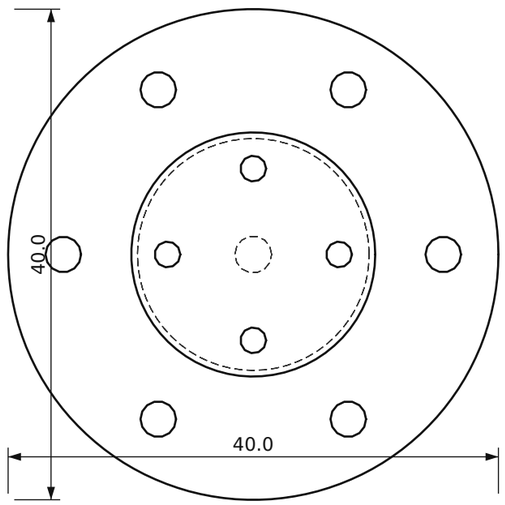


TOP

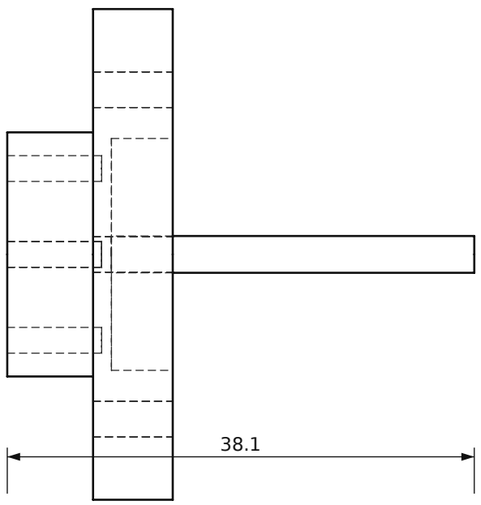


ISOMETRIC

HOLE	AXIS	QTY
Ø40	Y	1
Ø18.9	Y	1
Ø2.9	Y	6



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT FLANGE

output_flange (J4)

PART / DWG No. P-030
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

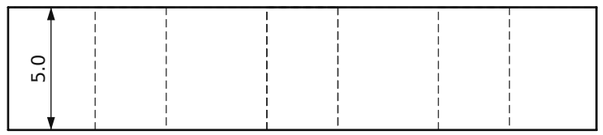
CHESS ROBOT ARM

SHEET 30 / 51

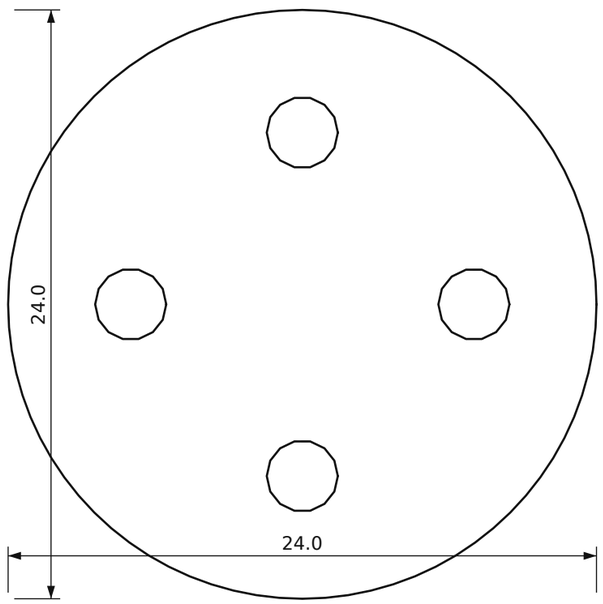
2026-11-12

ALL DIMENSIONS IN MM

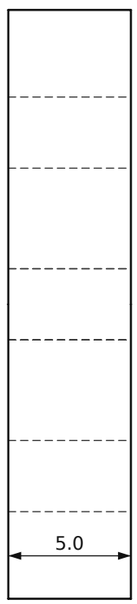
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



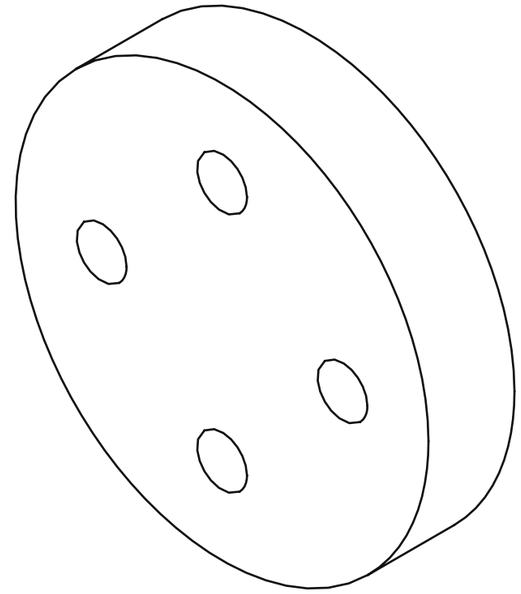
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø24	Y	1
Ø2.9	Y	4

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

OUTPUT CLAMP

output_clamp (J4)

PART / DWG No. P-031
MATERIAL PETG

SCALE	QTY	UNITS
4:1	**1**	**mm**

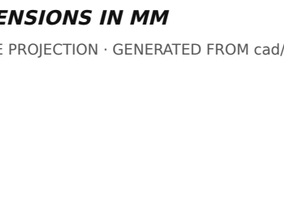
CHESS ROBOT ARM

SHEET 31 / 51

2026-11-12

ALL DIMENSIONS IN MM

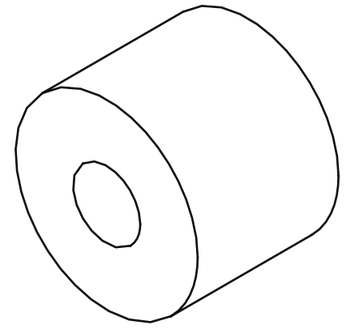
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



TOP

FRONT

RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø8.4	Y	1
Ø6.1	Y	1

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

MAGNET CAP

magnet_cap (J4)

PART / DWG No. P-032
MATERIAL PETG

SCALE	QTY	UNITS
5:1	**1**	**mm**

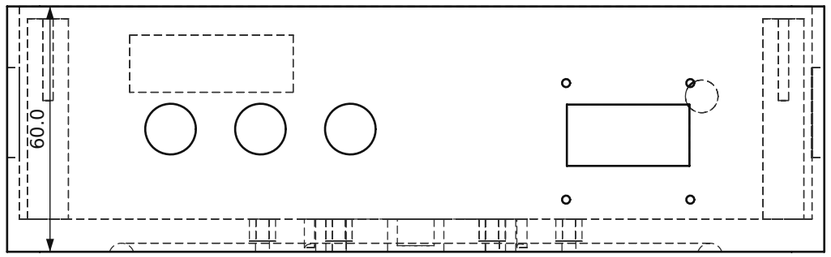
CHESS ROBOT ARM

SHEET 32 / 51

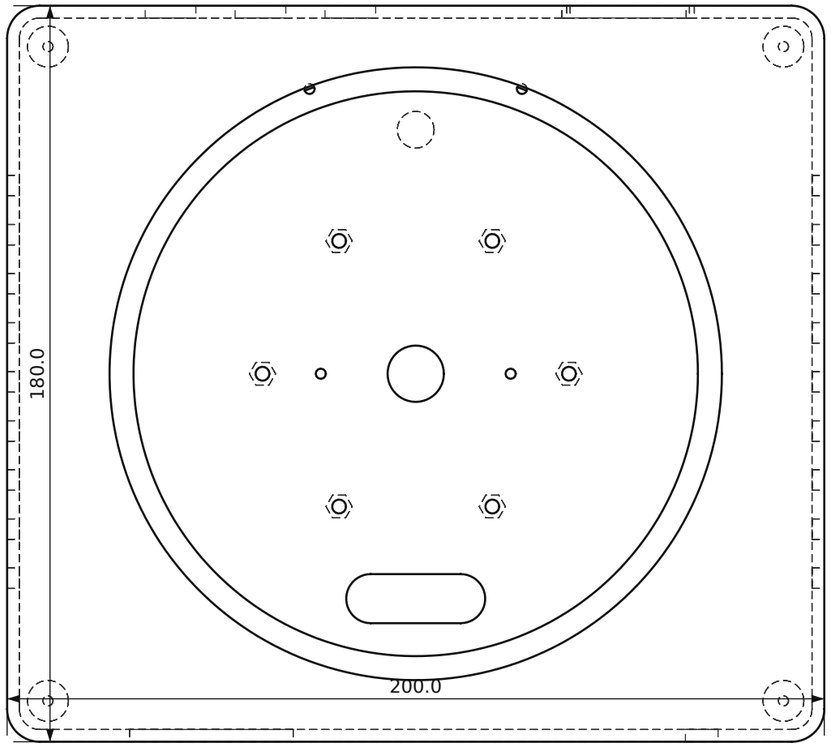
2026-11-12

ALL DIMENSIONS IN MM

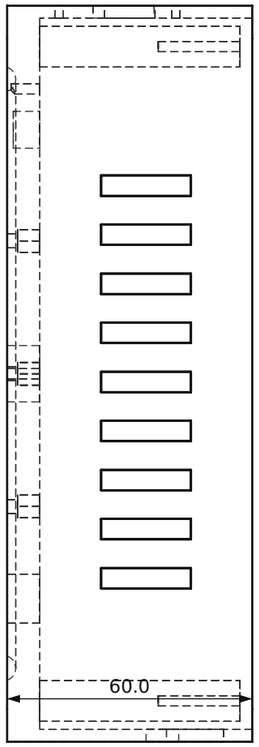
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



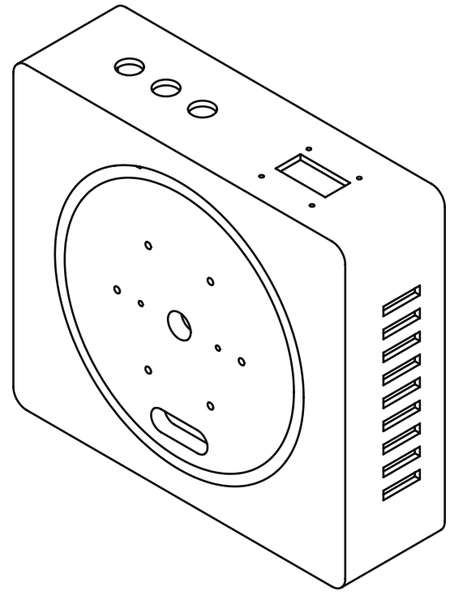
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø13.77	Y	1
Ø12.4	Z	3
Ø12	Y	2
Ø10	Y	8
Ø9	Y	1
Ø8	Z	1
Ø3.4	Y	6
Ø2.5	Y	8
Ø2	Z	4

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

BASE SHELL

Printed part

PART / DWG No. P-033
MATERIAL PETG

SCALE	QTY	UNITS
2:3	**1**	**mm**

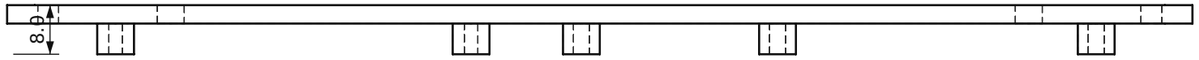
CHESS ROBOT ARM

SHEET 33 / 51

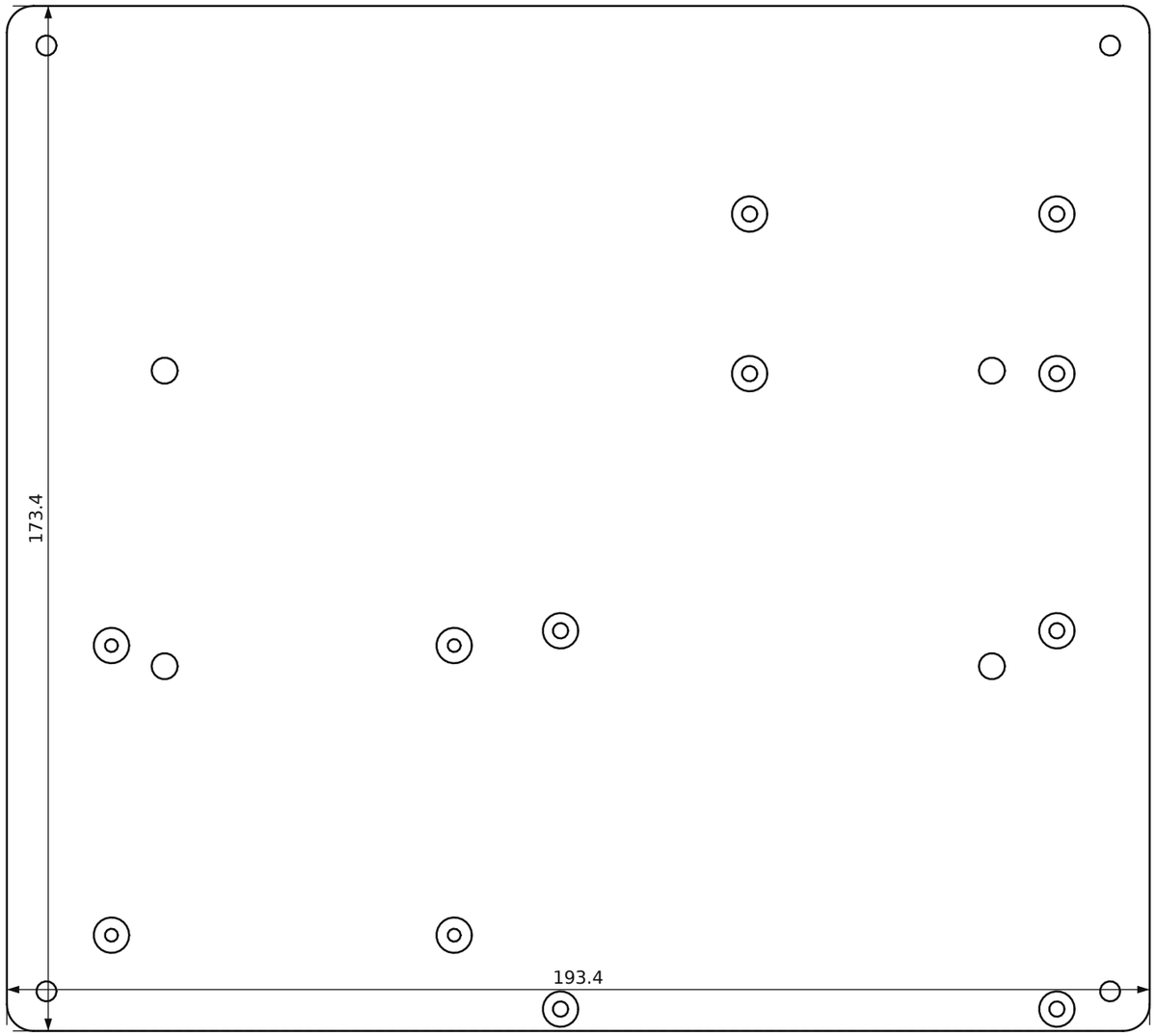
2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

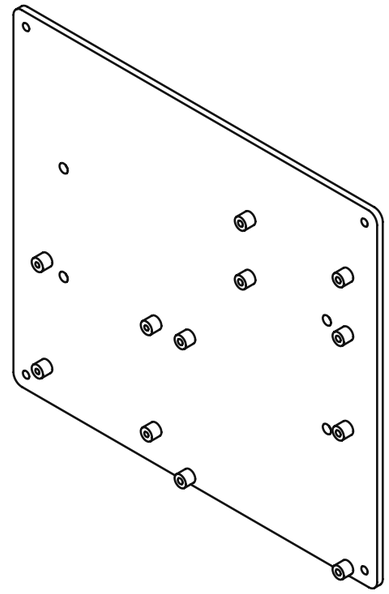


TOP



FRONT

RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø4.4	Y	4
Ø3.4	Y	4

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

BASE FLOOR

Printed part

PART / DWG No. P-034

MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

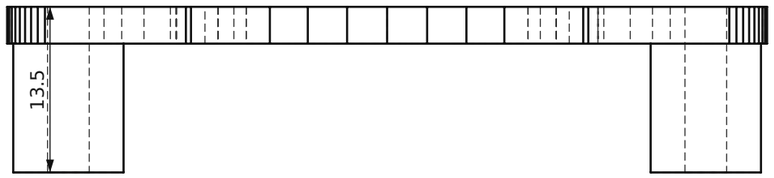
CHESS ROBOT ARM

SHEET 34 / 51

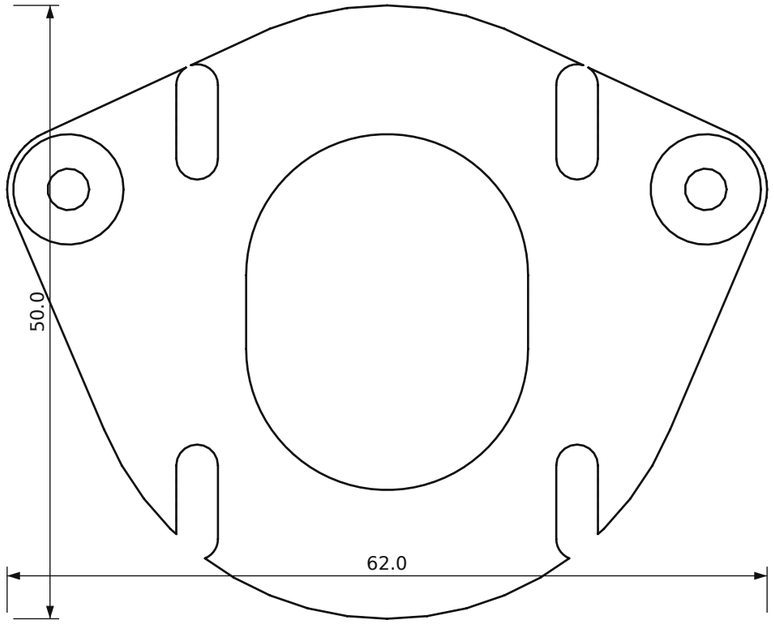
2026-11-12

ALL DIMENSIONS IN MM

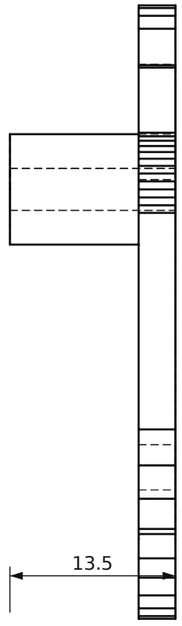
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



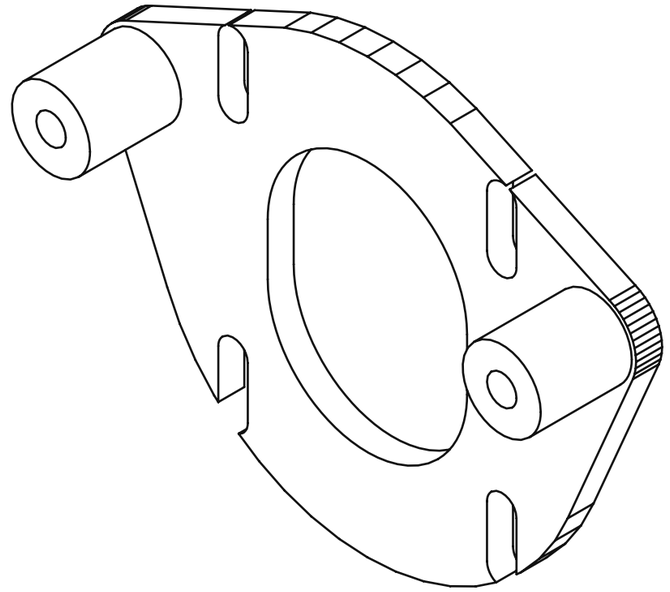
TOP



FRONT



RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø23	Y	2
Ø9	Y	2
Ø3.4	Y	10

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

J1 MOTOR SHELF

Printed part

PART / DWG No. P-035
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

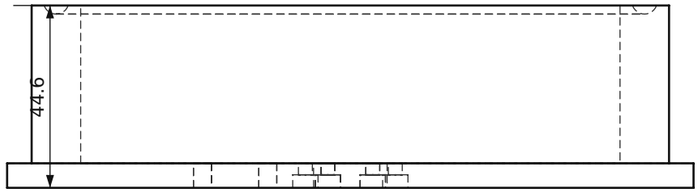
CHESS ROBOT ARM

SHEET 35 / 51

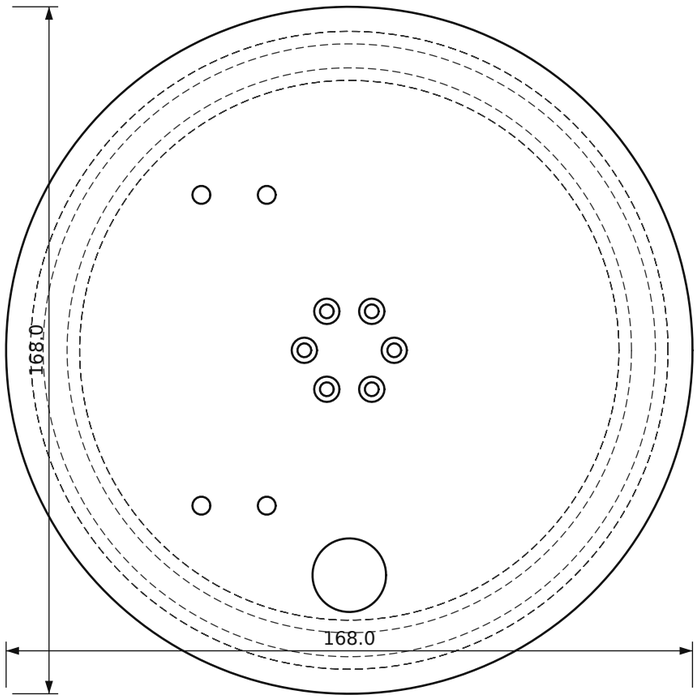
2026-11-12

ALL DIMENSIONS IN MM

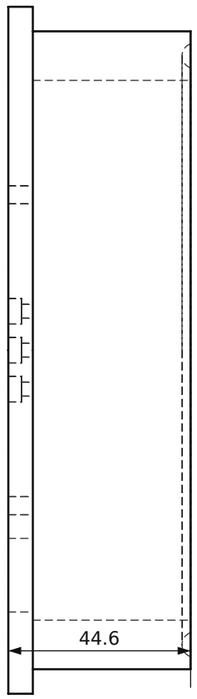
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



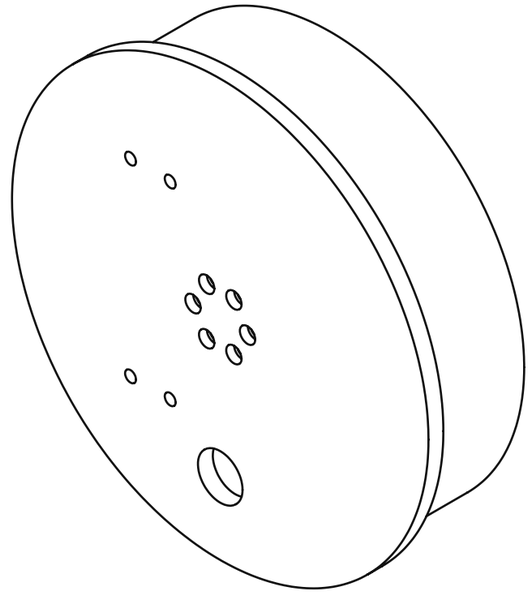
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø18	Y	1
Ø4.4	Y	4
Ø3.4	Y	6

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

TURNTABLE

Printed part

PART / DWG No. P-036
MATERIAL PETG

SCALE	QTY	UNITS
2:3	**1**	**mm**

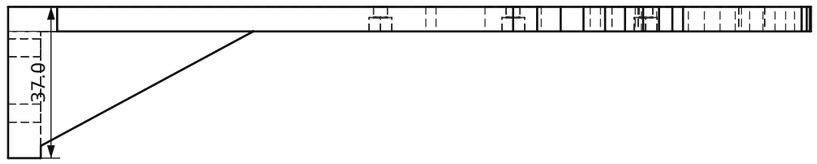
CHESS ROBOT ARM

SHEET 36 / 51

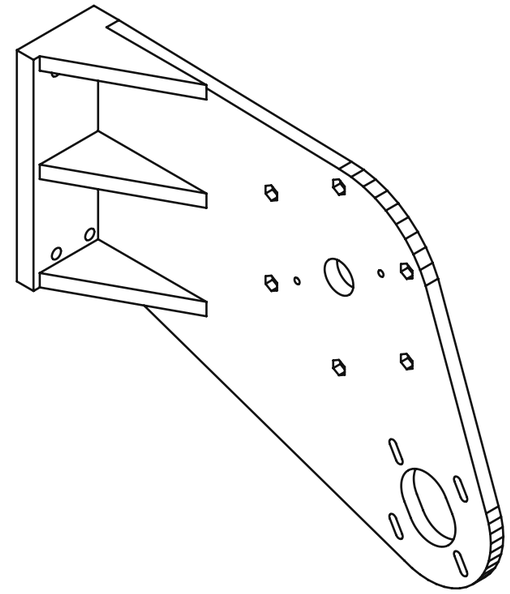
2026-11-12

ALL DIMENSIONS IN MM

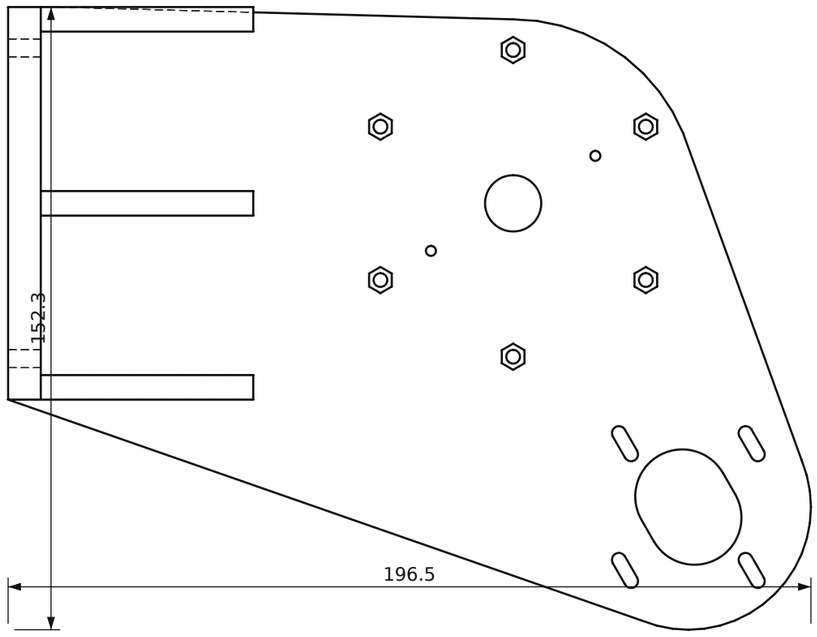
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



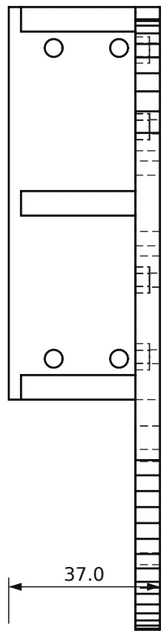
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

HOLE	AXIS	QTY
Ø23	Y	2
Ø13.77	Y	1
Ø4.4	X	4
Ø3.4	Y	14
Ø2.5	Y	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

TOWER

Printed part

PART / DWG No. P-037
MATERIAL PETG

SCALE	QTY	UNITS
2:3	**1**	**mm**

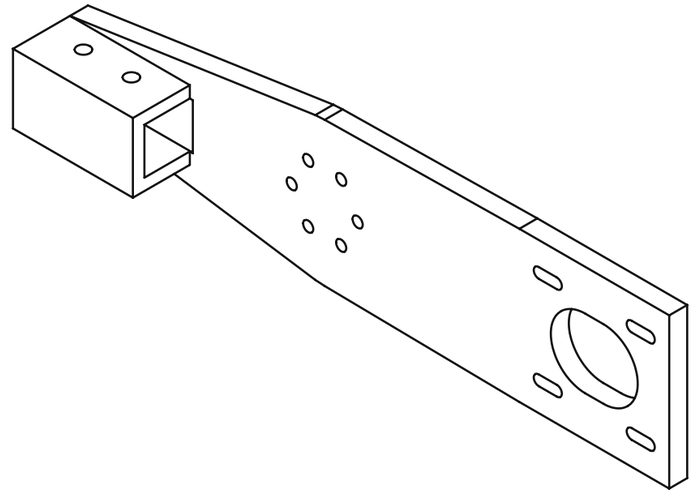
CHESS ROBOT ARM

SHEET 37 / 51

2026-11-12

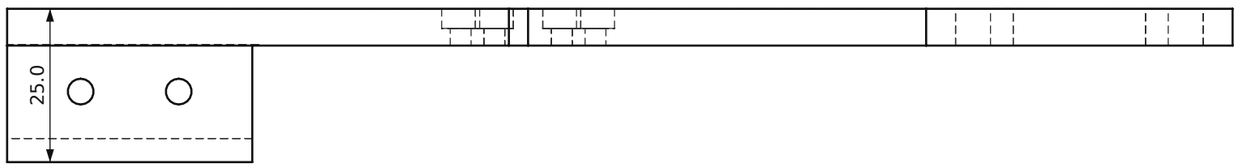
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

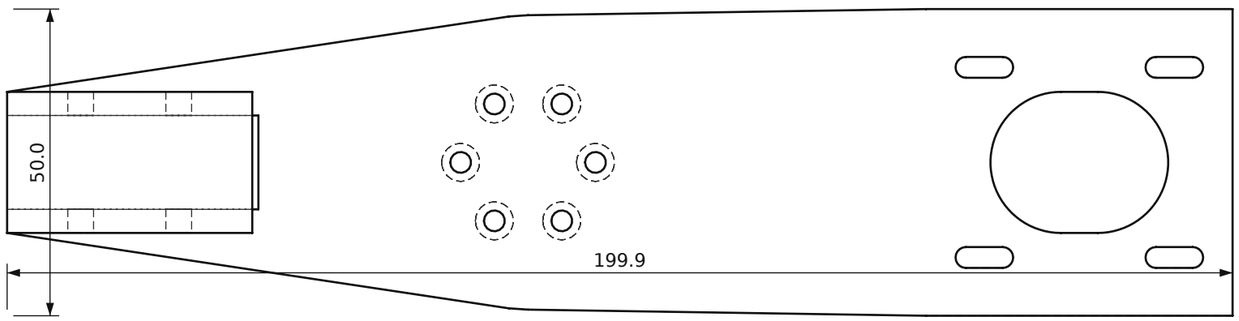


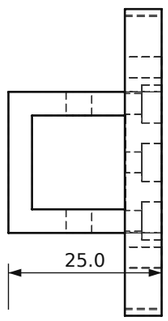
ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø23	Y	2
Ø4.2	Z	2
Ø3.4	Y	14



TOP





FRONT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

UPPER PROXIMAL

Printed part **RIGHT**

PART / DWG No. P-038
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

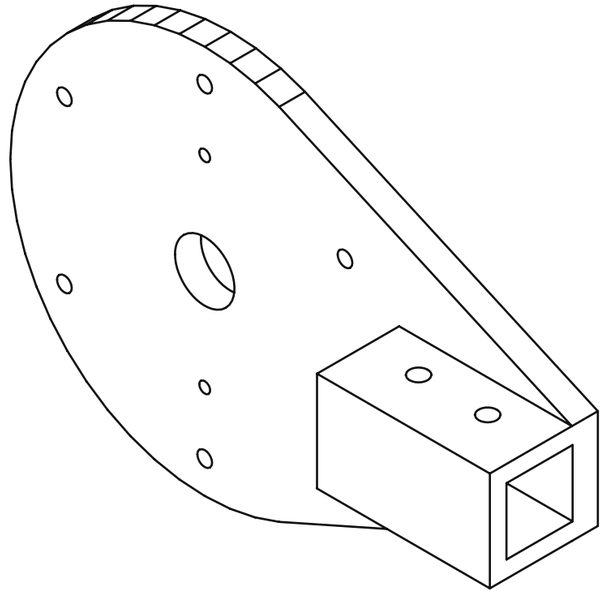
CHESS ROBOT ARM

SHEET 38 / 51

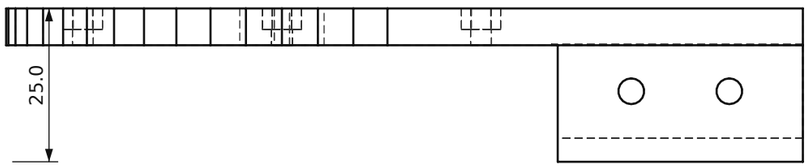
2026-11-12

ALL DIMENSIONS IN MM

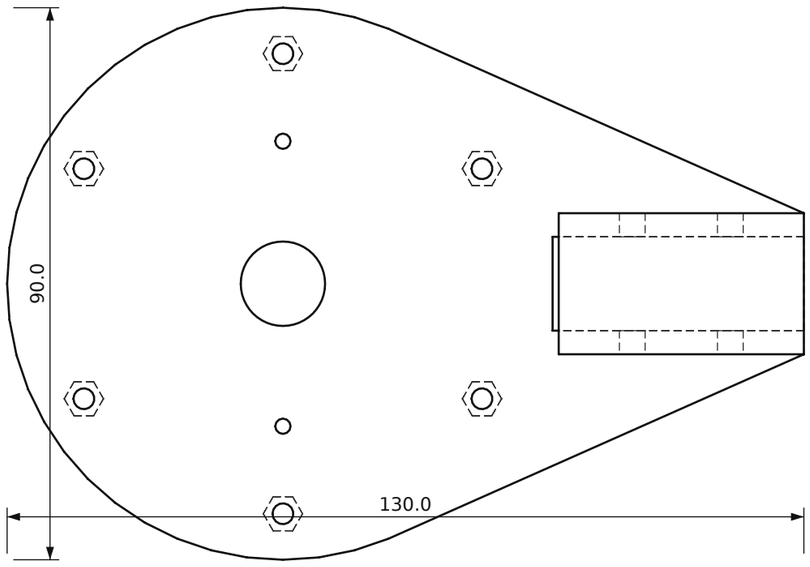
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



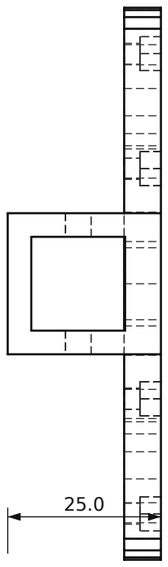
ISOMETRIC (NOT TO SCALE)



TOP



FRONT



RIGHT

HOLE	AXIS	QTY
Ø13.77	Y	1
Ø4.2	Z	2
Ø3.4	Y	6
Ø2.5	Y	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

UPPER DISTAL

Printed part

PART / DWG No. P-039
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

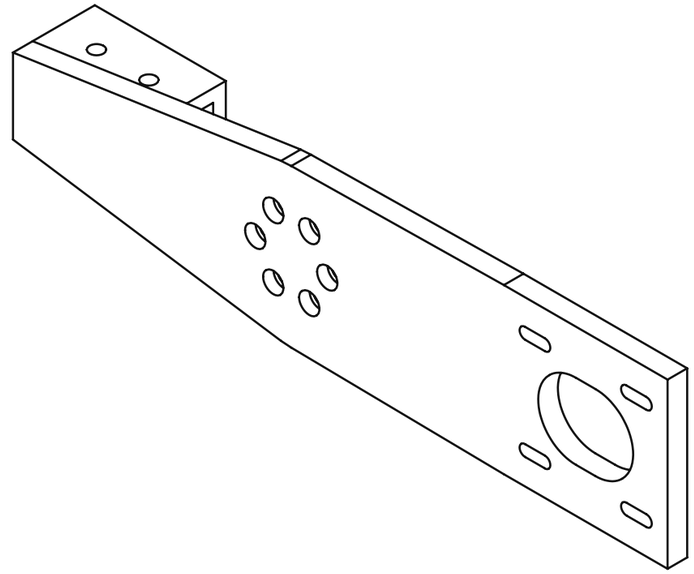
CHESS ROBOT ARM

SHEET 39 / 51

2026-11-12

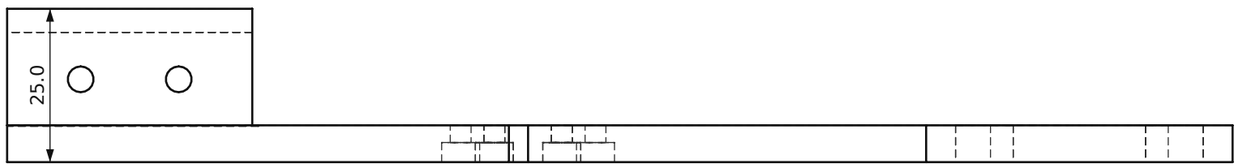
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

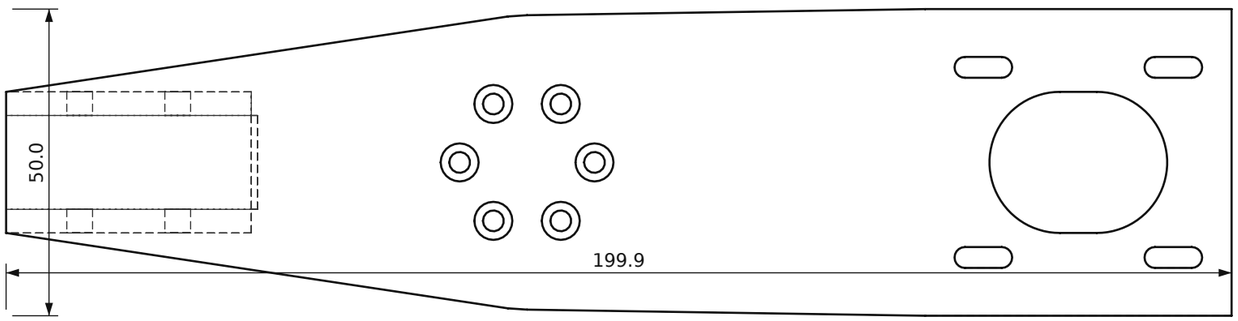


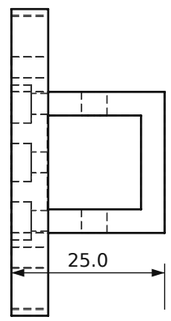
ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø23	Y	2
Ø6.2	Y	6
Ø4.2	Z	2
Ø3.4	Y	14



TOP





FRONT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

FORE PROXIMAL

Printed part **RIGHT**

PART / DWG No. P-040
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

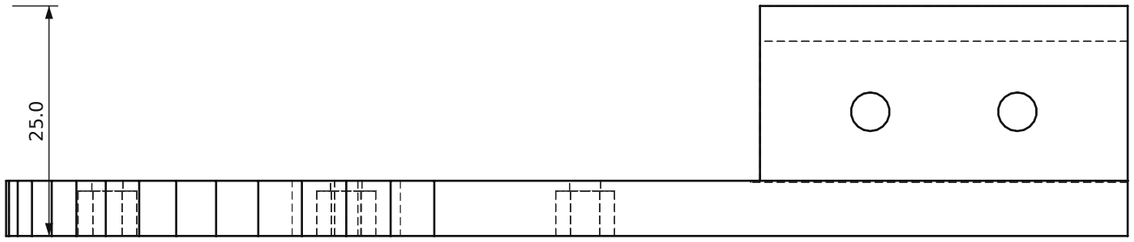
CHESS ROBOT ARM

SHEET 40 / 51

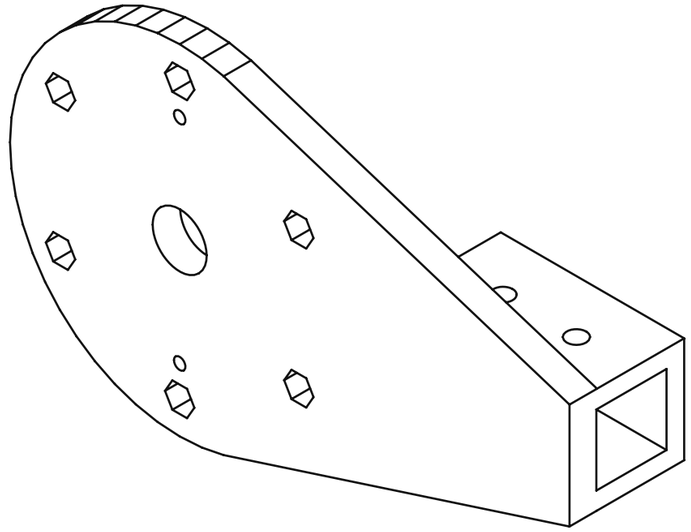
2026-11-12

ALL DIMENSIONS IN MM

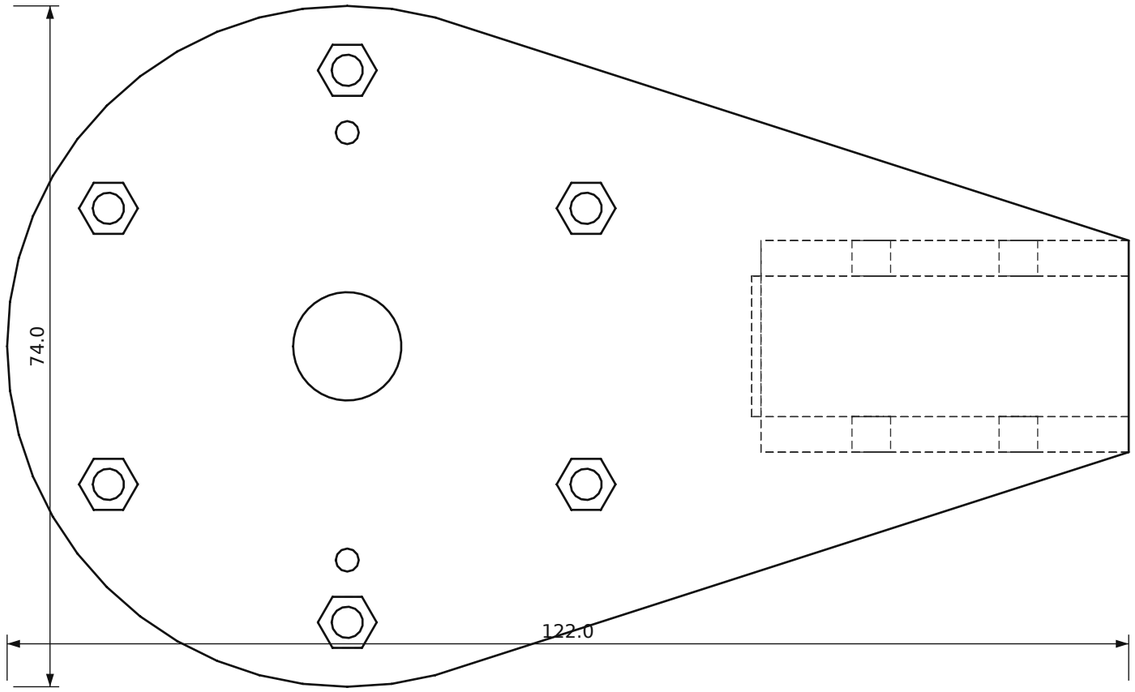
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



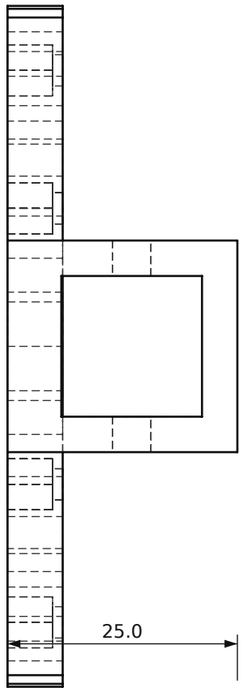
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



HOLE	AXIS	QTY
Ø11.77	Y	1
Ø4.2	Z	2
Ø3.4	Y	6
Ø2.5	Y	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

FORE DISTAL

Printed part **RIGHT**

PART / DWG No. P-041
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

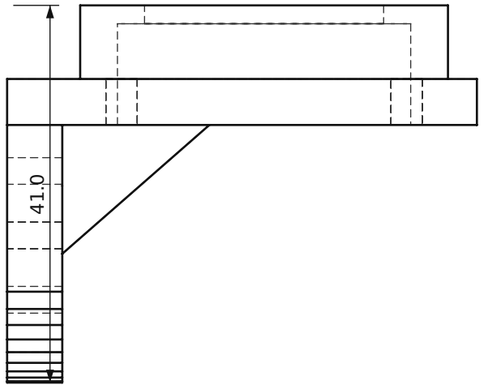
CHESS ROBOT ARM

SHEET 41 / 51

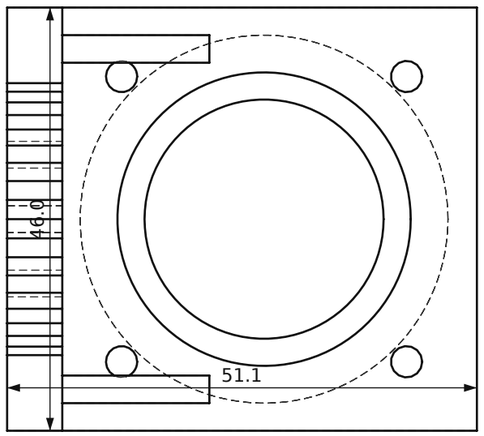
2026-11-12

ALL DIMENSIONS IN MM

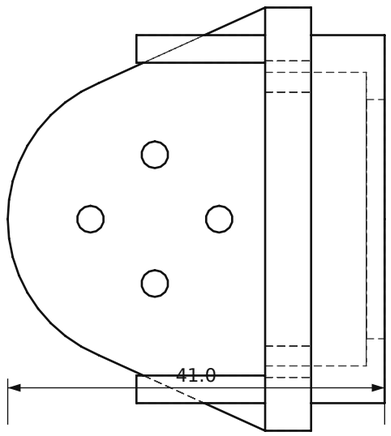
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



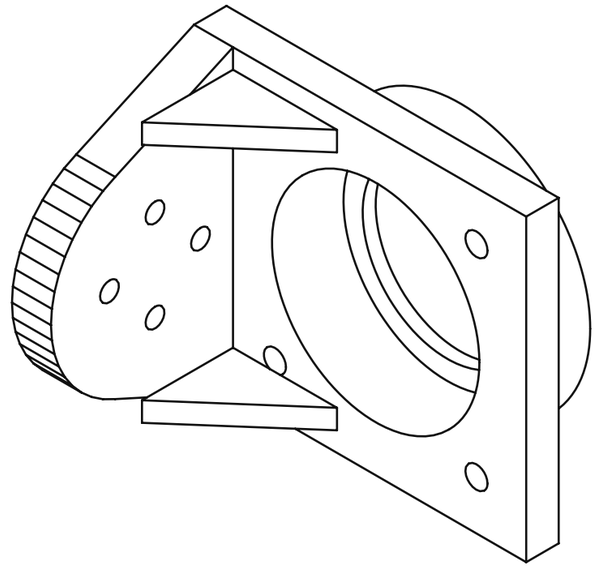
TOP



FRONT



RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø40	Y	1
Ø26	Y	1
Ø3.4	Y	4
Ø2.9	X	4

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

WRIST BRACKET

Printed part

PART / DWG No. P-042
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

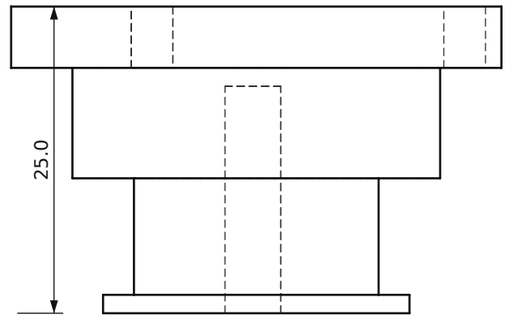
CHESS ROBOT ARM

SHEET 42 / 51

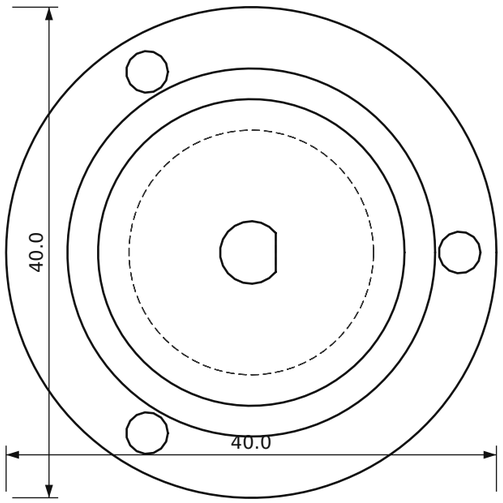
2026-11-12

ALL DIMENSIONS IN MM

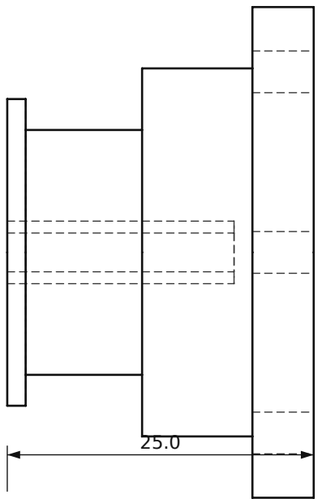
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



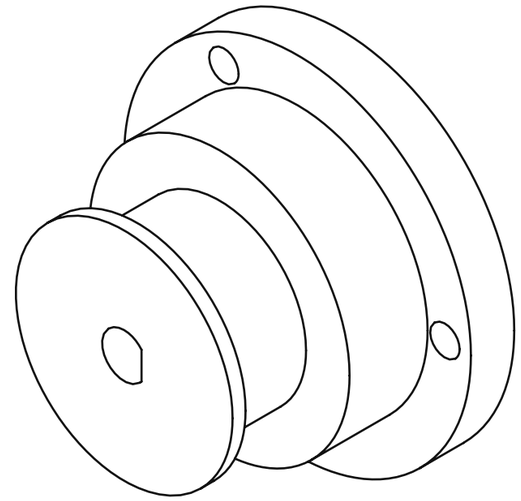
TOP



FRONT



RIGHT



ISOMETRIC

HOLE	AXIS	QTY
Ø40	Y	1
Ø30	Y	1
Ø25	Y	1
Ø3.4	Y	3

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

ROLL HUB

Printed part

PART / DWG No. P-043
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

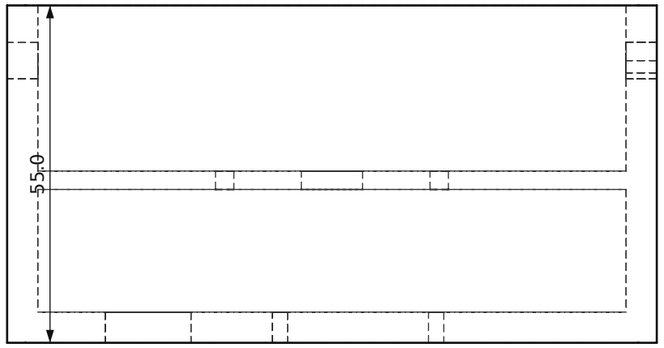
CHESS ROBOT ARM

SHEET 43 / 51

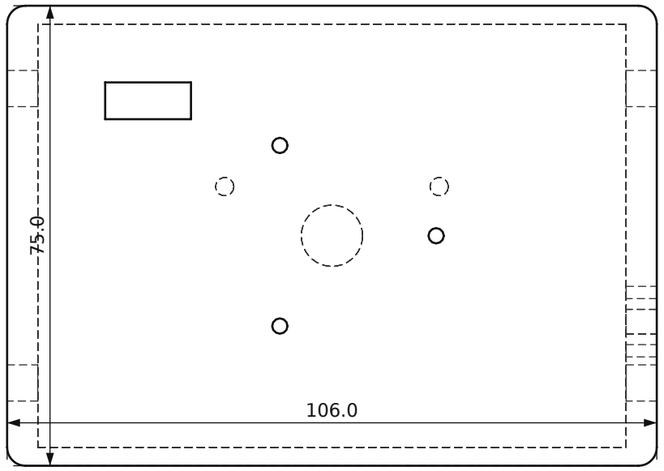
2026-11-12

ALL DIMENSIONS IN MM

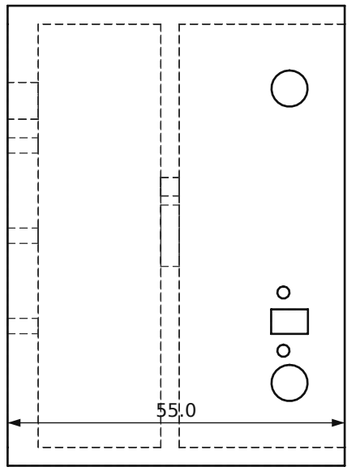
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



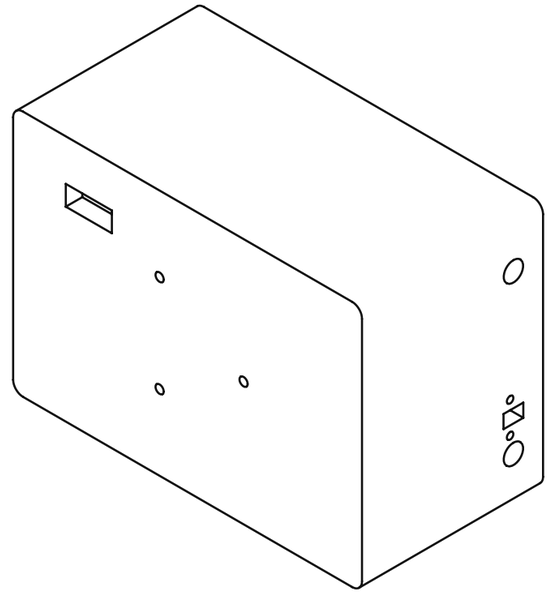
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø10	Y	1
Ø5.9	X	2
Ø3	Y	2
Ø2.5	Y	3
Ø2	X	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

GRIPPER FRAME

Printed part

PART / DWG No. P-044
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

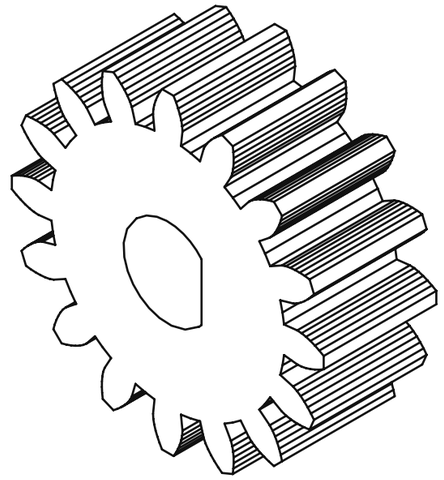
CHESS ROBOT ARM

SHEET 44 / 51

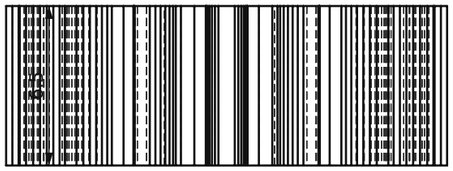
2026-11-12

ALL DIMENSIONS IN MM

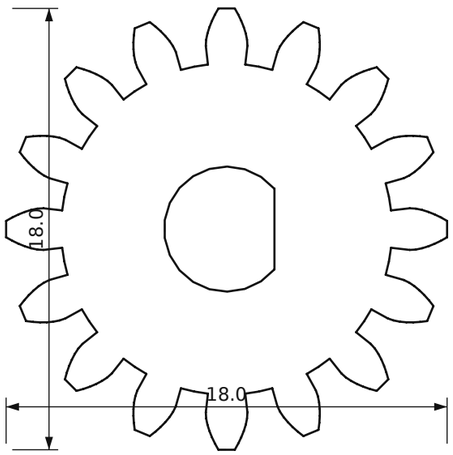
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



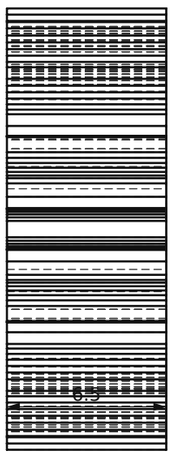
ISOMETRIC



TOP



FRONT



RIGHT

HOLE	AXIS	QTY
Ø5.1	Y	1

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

PINION

Printed part

PART / DWG No. P-045
MATERIAL PETG

SCALE	QTY	UNITS
4:1	**1**	**mm**

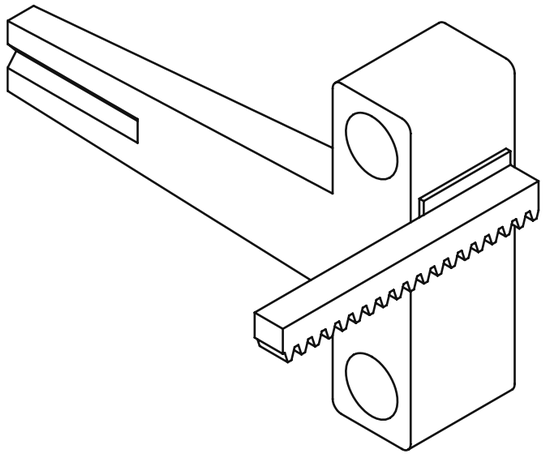
CHESS ROBOT ARM

SHEET 45 / 51

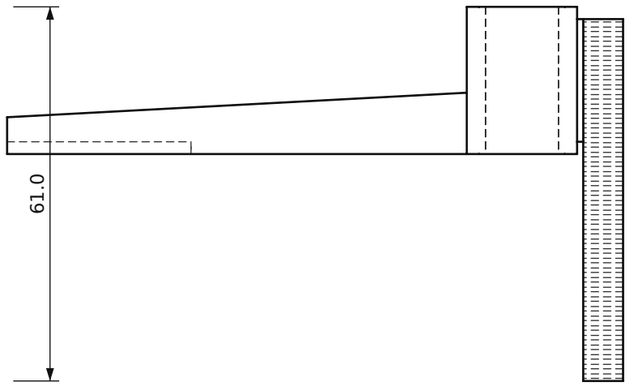
2026-11-12

ALL DIMENSIONS IN MM

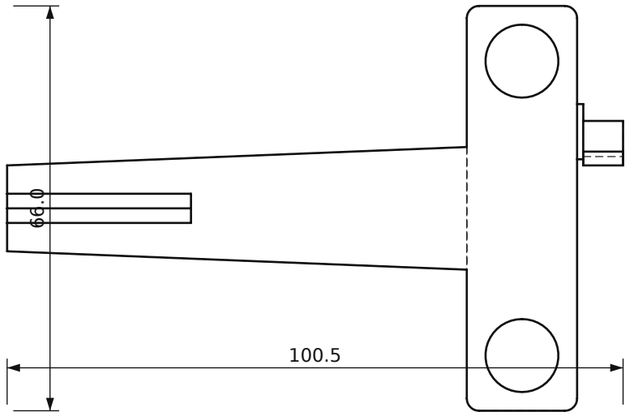
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



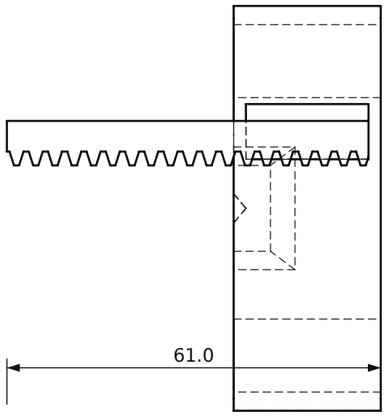
ISOMETRIC



TOP



FRONT



RIGHT

HOLE	AXIS	QTY
Ø11.9	Y	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

JAW A

Printed part

PART / DWG No. P-046
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

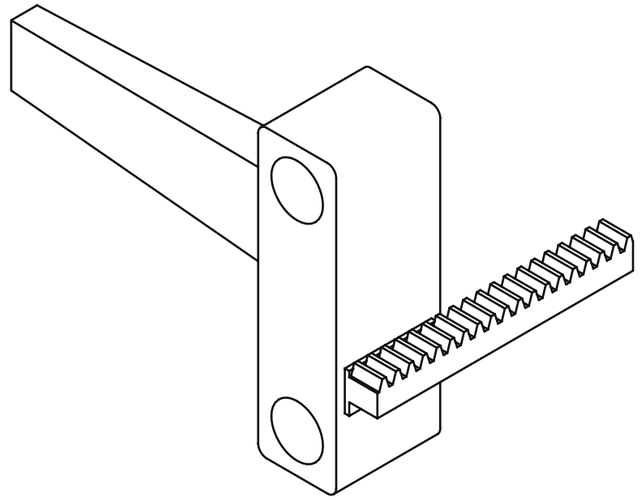
CHESS ROBOT ARM

SHEET 46 / 51

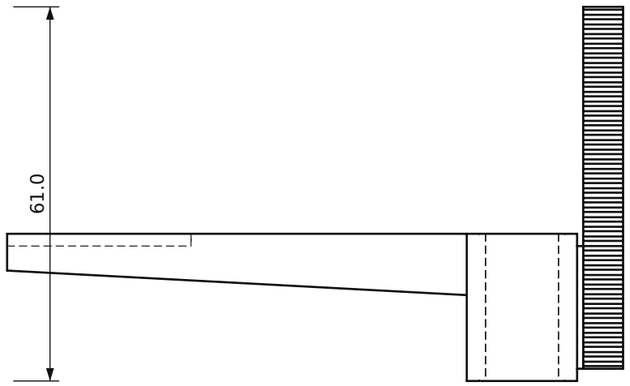
2026-11-12

ALL DIMENSIONS IN MM

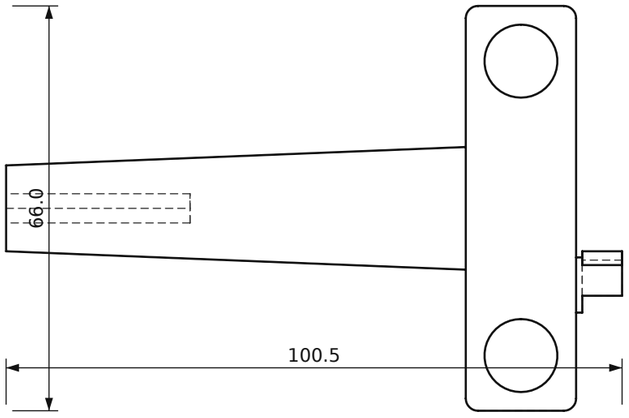
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



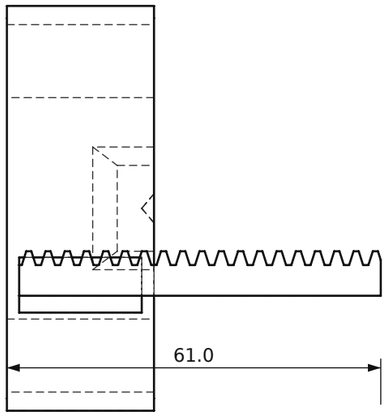
ISOMETRIC



TOP



FRONT



RIGHT

HOLE	AXIS	QTY
Ø11.9	Y	2

PRINT: PETG, 0.2 mm layers, 4 perimeters, 100 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

JAW B

Printed part

PART / DWG No. P-047
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

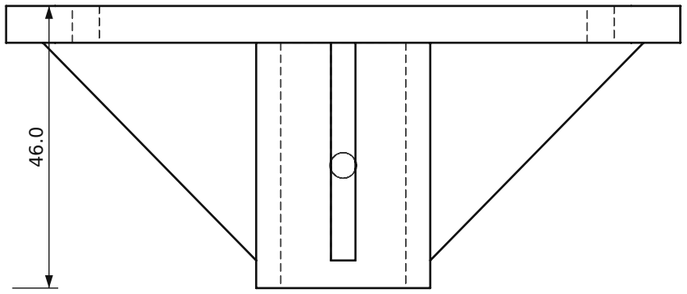
CHESS ROBOT ARM

SHEET 47 / 51

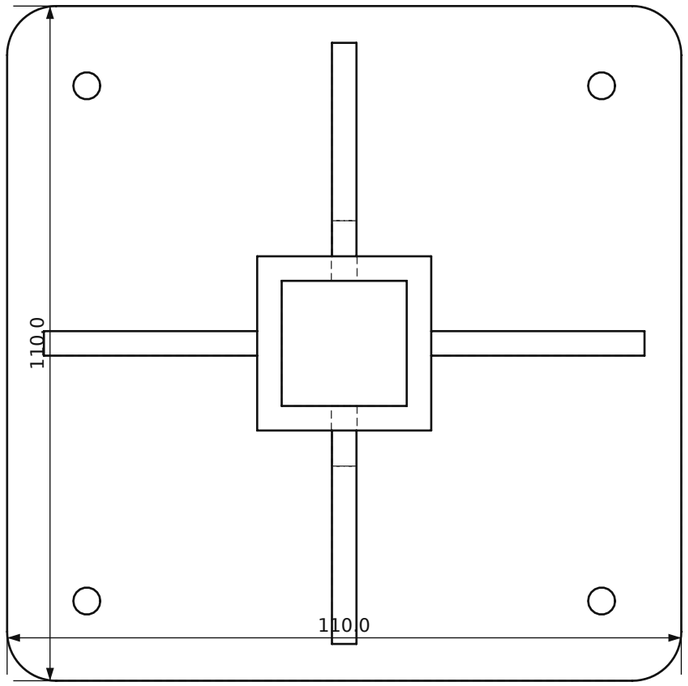
2026-11-12

ALL DIMENSIONS IN MM

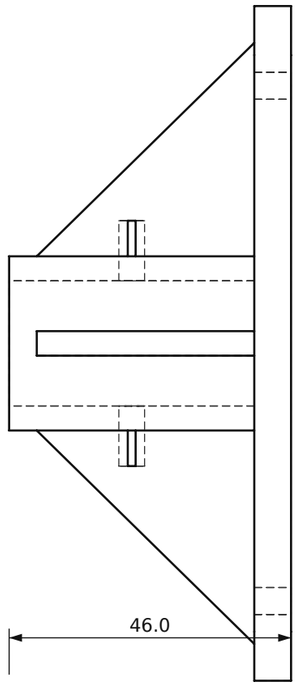
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



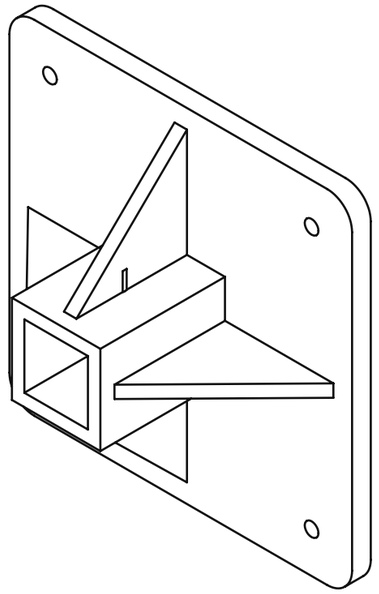
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø4.4	Y	4

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CAMERA FOOT

Printed part

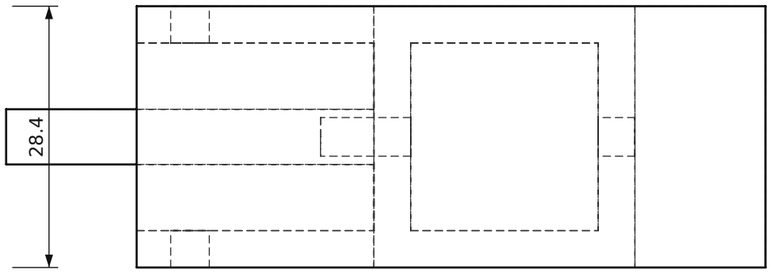
PART / DWG No. P-048
MATERIAL PETG

SCALE	QTY	UNITS
1:1	**1**	**mm**

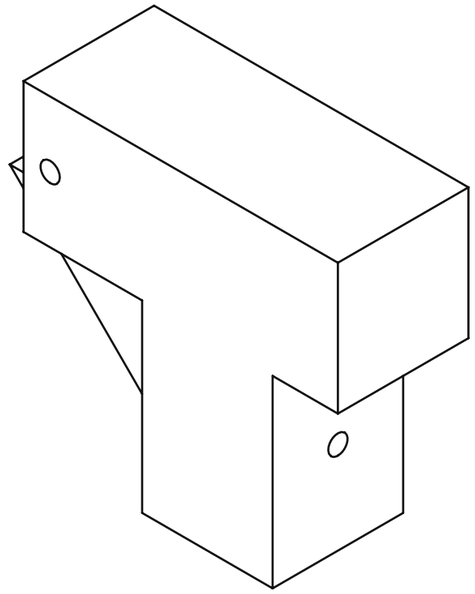
CHESS ROBOT ARM
SHEET 48 / 51
2026-11-12

ALL DIMENSIONS IN MM

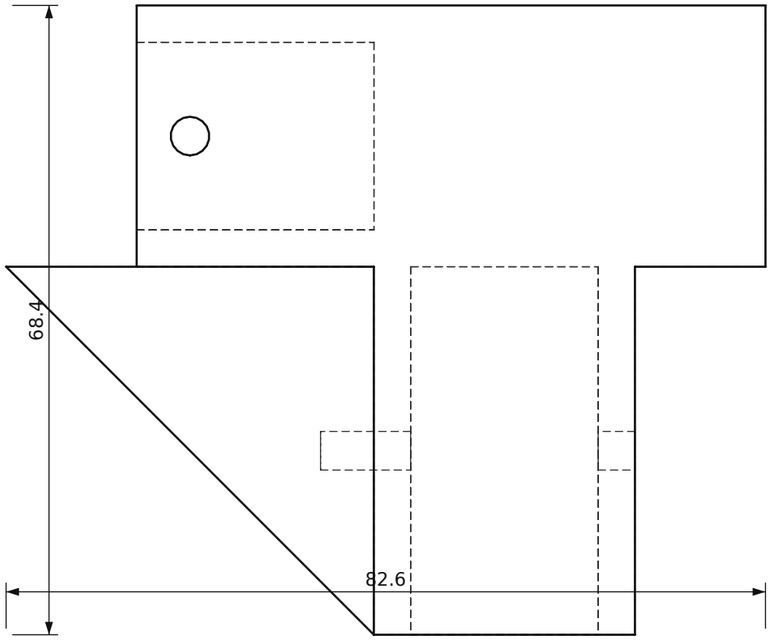
THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)



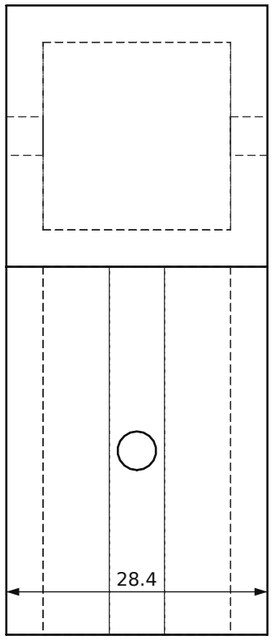
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

HOLE	AXIS	QTY
Ø4.2	Y	1

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CAMERA TEE

Printed part

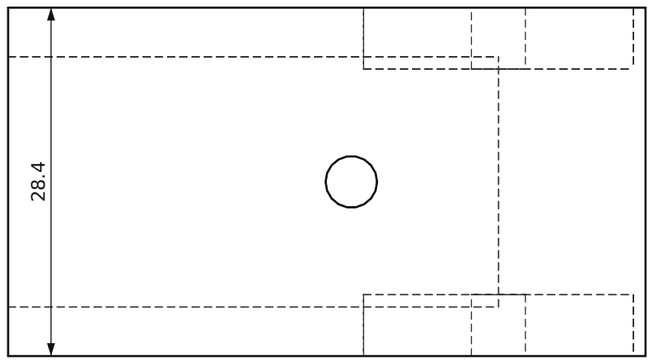
PART / DWG No. P-049
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

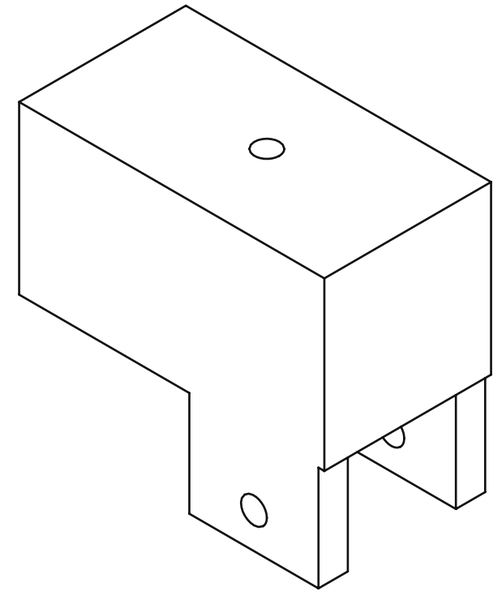
CHESS ROBOT ARM
SHEET 49 / 51
2026-11-12

ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

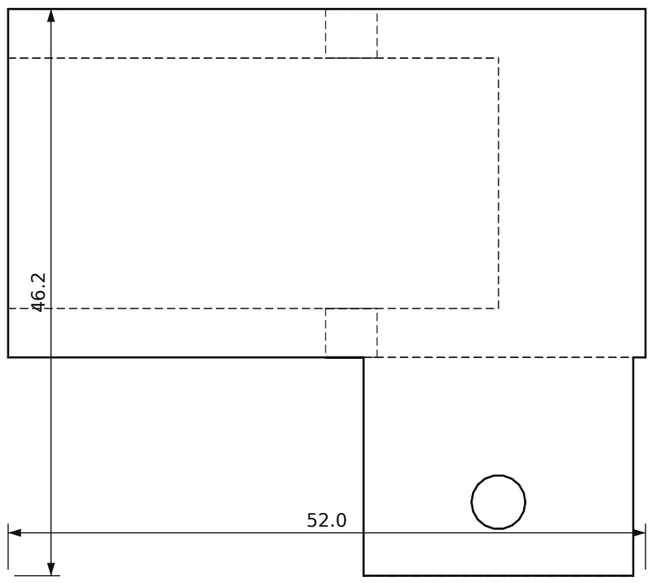


TOP

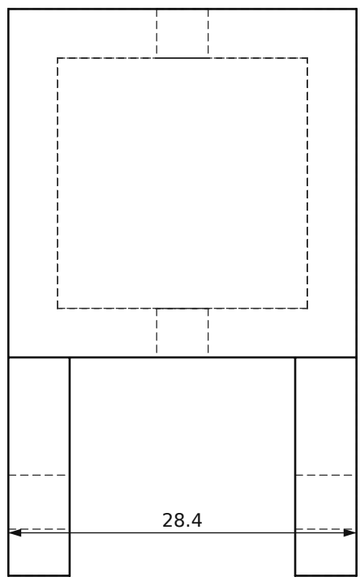


ISOMETRIC (NOT TO SCALE)

HOLE	AXIS	QTY
Ø4.4	Y	1
Ø4.2	Z	1



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CAMERA MOUNT

Printed part

PART / DWG No. P-050
MATERIAL PETG

SCALE	QTY	UNITS
2:1	**1**	**mm**

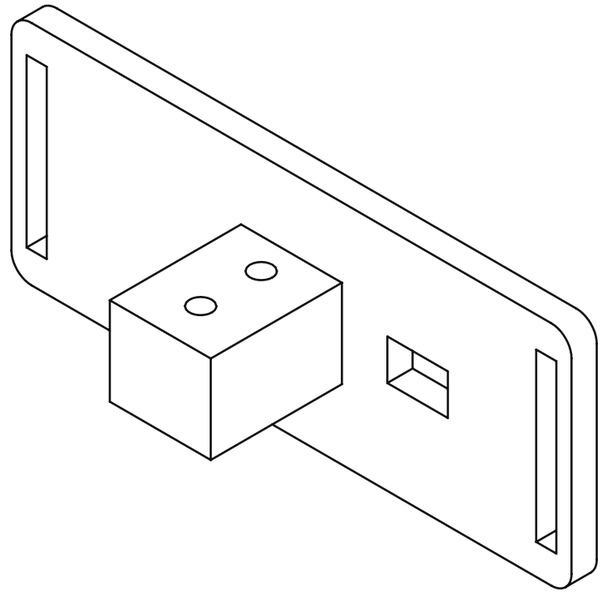
CHESS ROBOT ARM

SHEET 50 / 51

2026-11-12

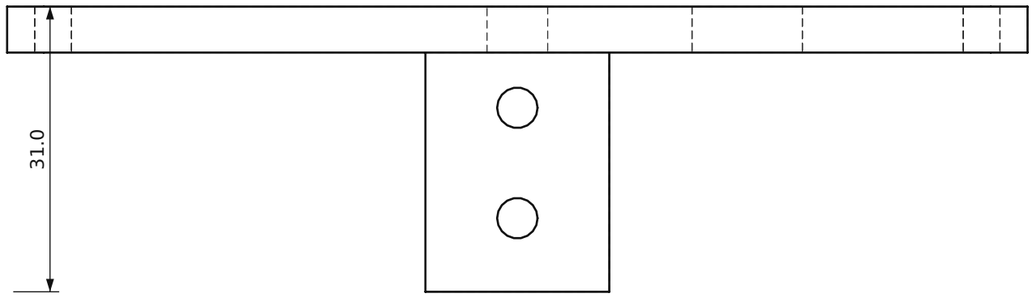
ALL DIMENSIONS IN MM

THIRD-ANGLE PROJECTION · GENERATED FROM cad/ (CadQuery)

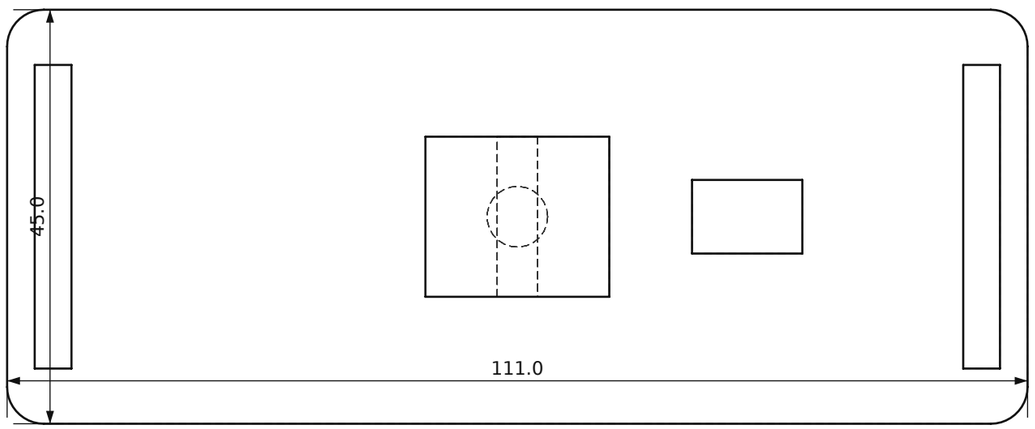


ISOMETRIC (NOT TO SCALE)

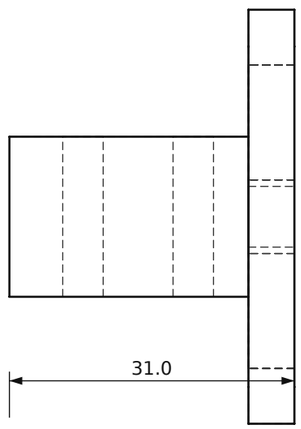
HOLE	AXIS	QTY
Ø6.6	Y	1
Ø4.4	Z	2



TOP



FRONT



RIGHT

PRINT: PETG, 0.2 mm layers, 4 perimeters, 40 % infill.
HOLES ARE MODELLED AT PRINT SIZE (FIT ALLOWANCES INCLUDED); BEARING SEATS PRESS-FIT.
UNTOLERANCED: ±0.2 mm. BREAK SHARP EDGES.

CAMERA PLATE

Printed part

PART / DWG No. P-051
MATERIAL PETG

SCALE	QTY	UNITS
3:2	**1**	**mm**

CHESS ROBOT ARM

SHEET 51 / 51

2026-11-12